

Unit 7

WAVES

45 items: 23 from official papers (83 marks), 22 from the booklet

Syllabus sections in this unit

- 7.1 Progressive waves
- 7.2 Transverse and longitudinal waves
- 7.3 Doppler effect for sound waves
- 7.4 Electromagnetic spectrum
- 7.5 Polarisation

Unit 7: Waves**1. M/J 26/P22/Q3/a,b(i)**

- 3** Two oscillating dippers are used to create progressive transverse water waves on the surface of a lake. The waves are shown from above in Figure 3.1.

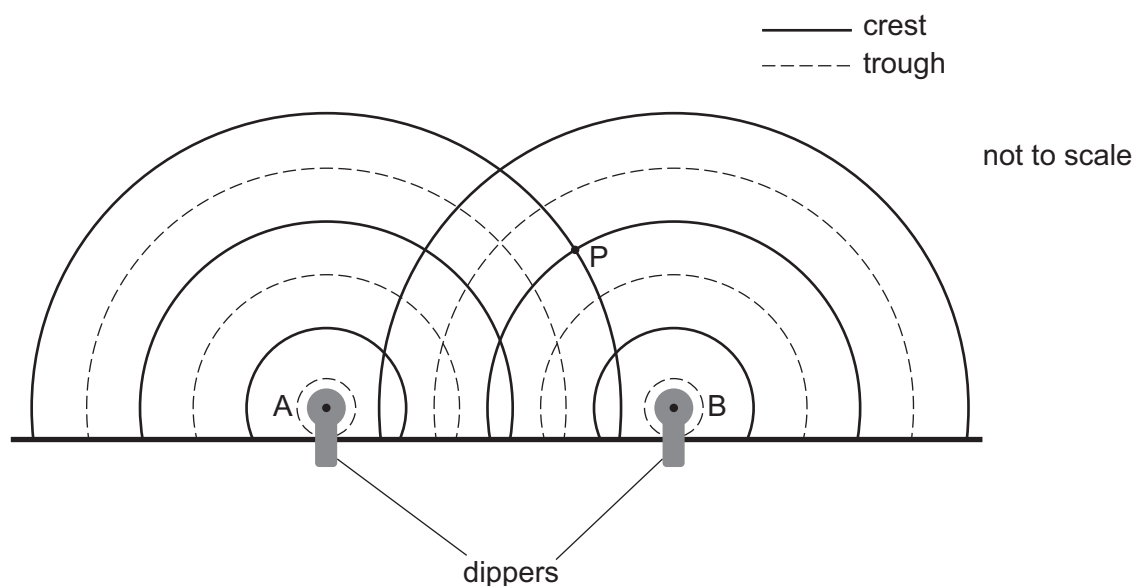


Figure 3.1

Wave crests are represented by a solid line. Wave troughs are represented by a dashed line. The waves superpose and an interference pattern is formed on the surface of the water.

The dippers oscillate in phase with each other and with a frequency of 3.0 Hz.

- (a)** State what is meant by a progressive wave.

[1]

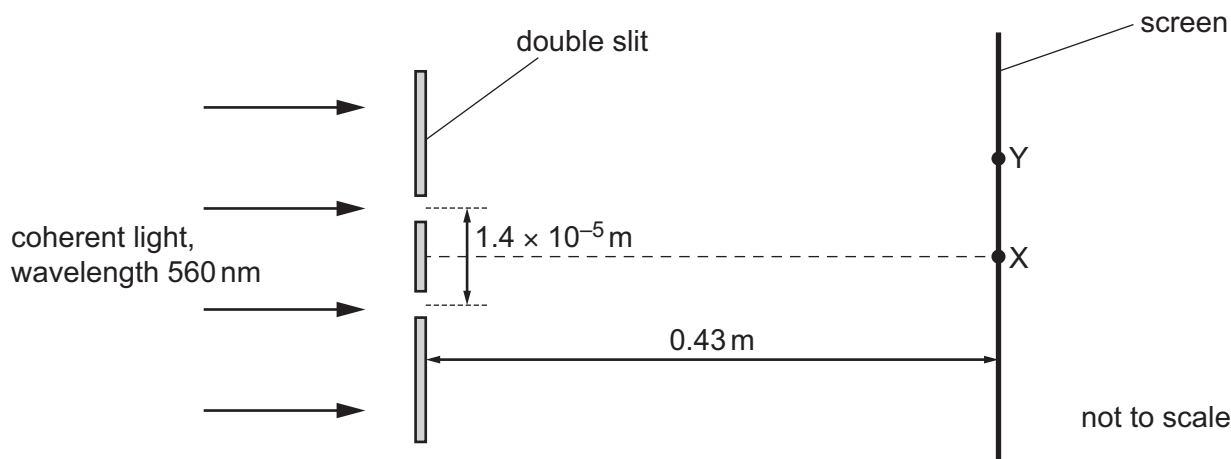
- (b) (i)** Calculate the period of the waves.

period =

s [1]

2. M/J 26/P23/Q3/d

- (b) Parallel rays of coherent light of wavelength 560 nm are incident normally in free space on two narrow slits as shown in Figure 3.1.

**Figure 3.1**

Light passing through the double slit reaches a screen that is parallel to the plane of the slits. The distance between the centres of the slits is $1.4 \times 10^{-5} \text{ m}$. The distance between the slits and the screen is 0.43 m.

- (i) State what is meant by coherent.

[1]

- (ii) State what happens to the light at the point where it passes through the slits.

[1]

- (iii) At point X on the screen, there is a central bright fringe.

Explain how this bright fringe is formed.

[3]

- (iv) At point Y on the screen, the dark fringe closest to point X is formed.

Calculate the distance XY.

distance = _____ m [3]

- (d) Calculate the frequency of the light in 3(b).

frequency = _____ Hz [2]

3. MAR 26/P22/Q5/a

- 5 (a) For a progressive wave, state what is meant by:

- (i) period

[1]

- (ii) amplitude.

[1]

4. O/N 25/P21/Q4/c

- (b) A beam of vertically polarised light of wavelength 540 nm is incident normally on a diffraction grating, as shown in Fig. 4.1.

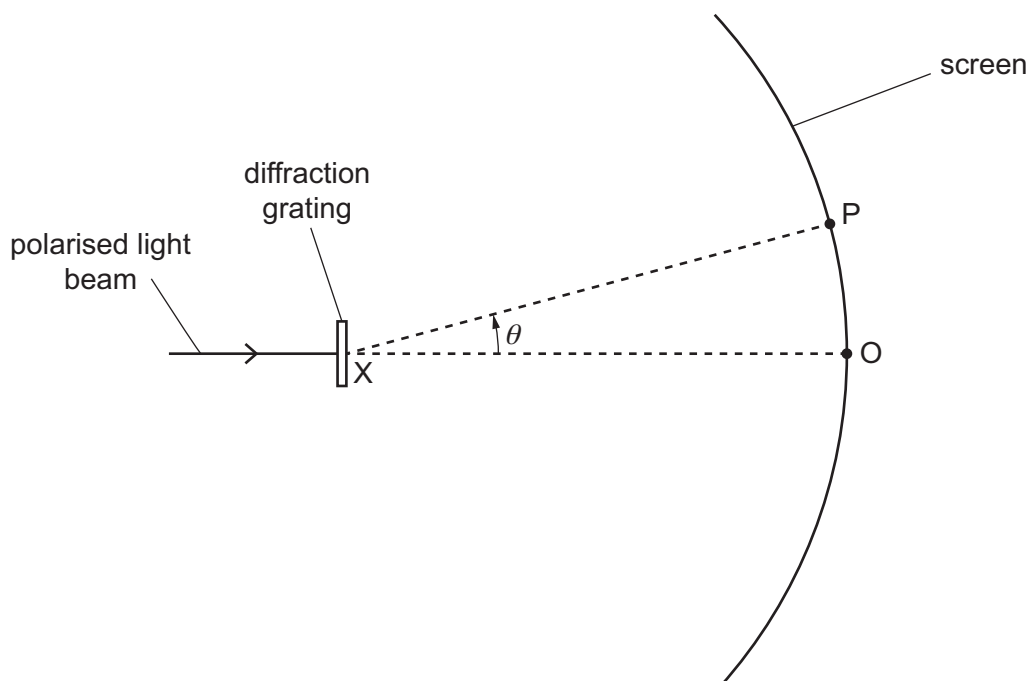


Fig. 4.1 (not to scale)

The diffraction grating has a line spacing of $5.0 \times 10^{-6} \text{ m}$.

The light transmitted by the diffraction grating illuminates a circular screen. The diffraction grating is at the centre X of the circle.

The central bright fringe is formed at point O on the screen and has intensity I_0 .

P is a point on the screen where the line XP is at a variable angle θ to the line XO. The intensity I of light on the screen at P varies with θ .

- (iii) On Fig. 4.2, sketch the variation of the intensity I with θ for values of θ from -15° to $+15^\circ$.

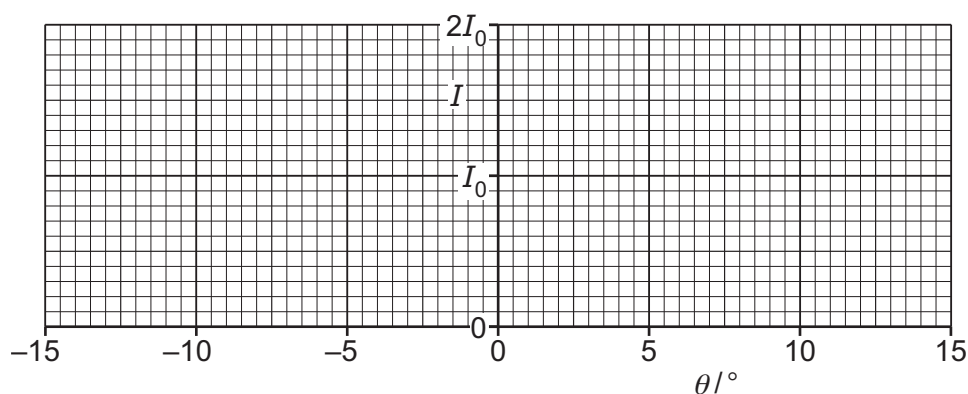


Fig. 4.2

[3]

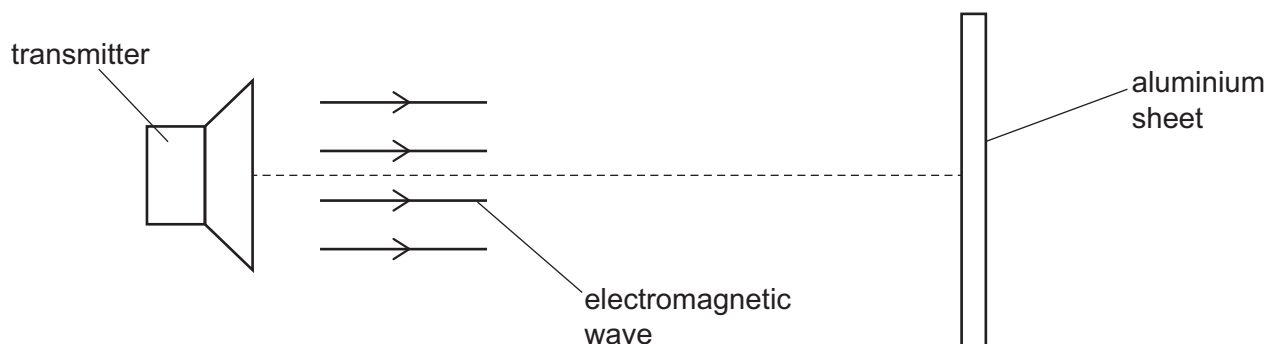
- (c) A polarising filter is placed in the path of the light beam that is incident on the diffraction grating in Fig. 4.1. The transmission axis of the filter is at 45° to the vertical.

Suggest how the variation of intensity with θ for the light on the screen compares with the answer in (b)(iii).

[2]

5. O/N 25/P24/Q4/b(ii,iii)

- (b) An electromagnetic wave of wavelength 0.026 m in free space is incident normally on an aluminium sheet, as shown in Fig. 4.1.

**Fig. 4.1**

The wave reflects at the aluminium sheet and a stationary wave is formed in the region between the transmitter and the sheet.

- (ii) Calculate the frequency of the electromagnetic wave.

frequency = _____ Hz [2]

- (iii) State the principal region of the electromagnetic spectrum to which the wave belongs.

[1]

6. M/J 25/P21/Q5/a,b

- 5 (a)** Use the definitions of speed v , frequency f and wavelength λ to derive the wave equation

$$v = f\lambda.$$

[2]

- (b)** A source of sound waves of frequency 236 Hz is travelling at a constant velocity of 20 m s^{-1} .

A stationary observer has a microphone connected to a cathode-ray oscilloscope (CRO). The microphone detects the sound waves as the source moves directly towards the observer.

The resulting trace on the CRO is shown in Fig. 5.1.

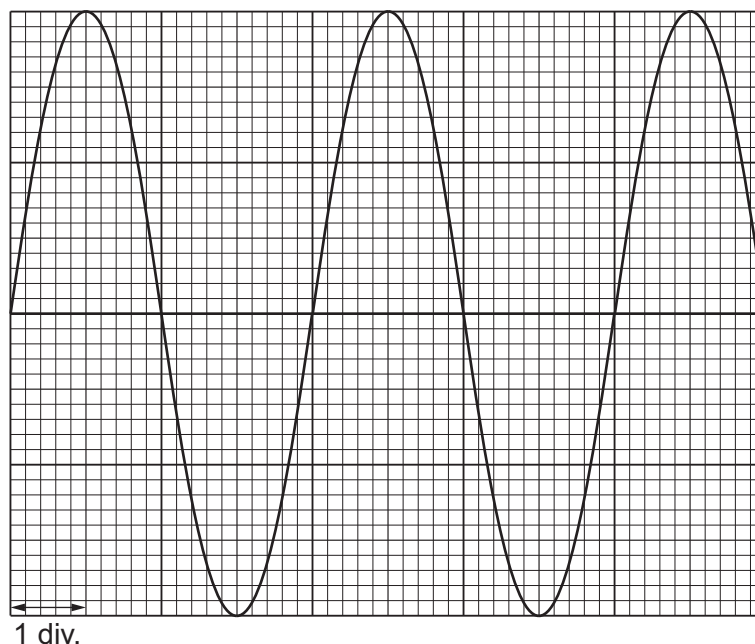


Fig. 5.1

The time-base on the CRO is set to 1.0 ms div^{-1} .

- (i)** Calculate the frequency of the sound waves detected by the microphone.

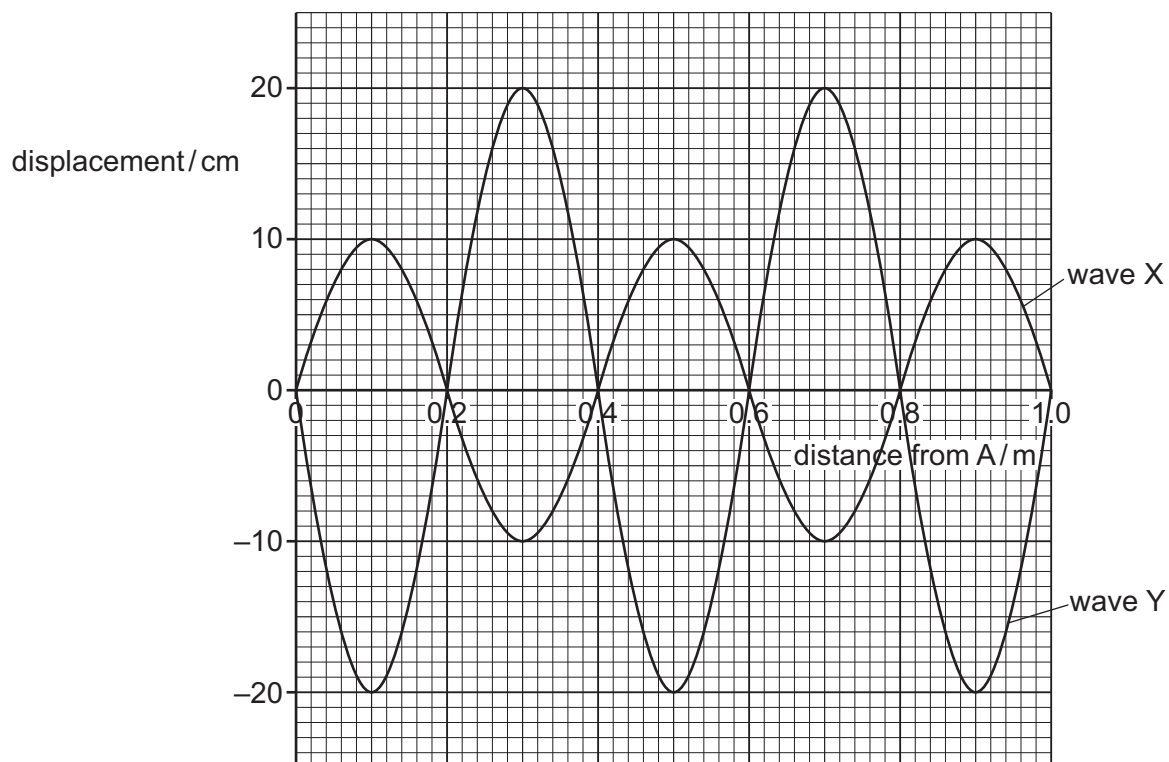
frequency = _____ Hz [2]

- (ii)** Determine the speed of the sound in air.

speed of sound = _____ ms^{-1} [2]

7. M/J 25/P22/Q3/a,b(i),d

- 3** Two progressive water waves X and Y travel along a straight line from point A to point B. The variation of displacement of the waves with distance from A at an instant in time is shown in Fig. 3.1.

**Fig. 3.1**

- (a)** State the amplitude of wave X.

amplitude = _____ cm [1]

- (b)** Both waves have frequency 16 Hz.

- (i)** Determine the speed of wave X.

speed = _____ ms^{-1} [2]

- (d)** The intensity of wave X is I_X . The intensity of wave Y is I_Y .

Use Fig. 3.1 to determine the ratio $\frac{I_X}{I_Y}$.

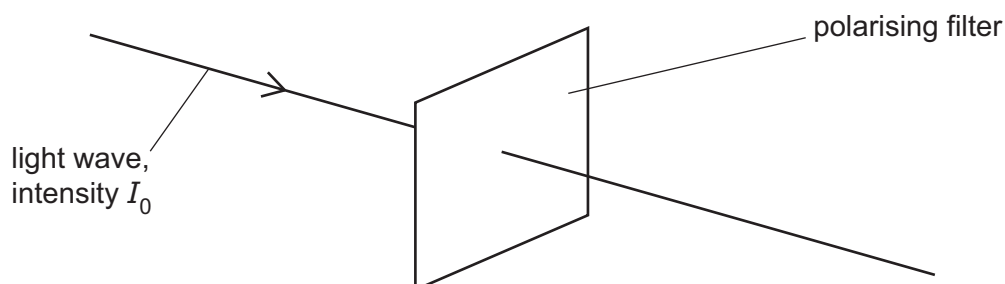
ratio = _____ [2]

8. M/J 25/P23/Q5/a,b

- 5 (a) State why sound waves **cannot** be polarised.

[1]

- (b) A plane-polarised light wave is incident on a polarising filter as shown in Fig. 5.1.

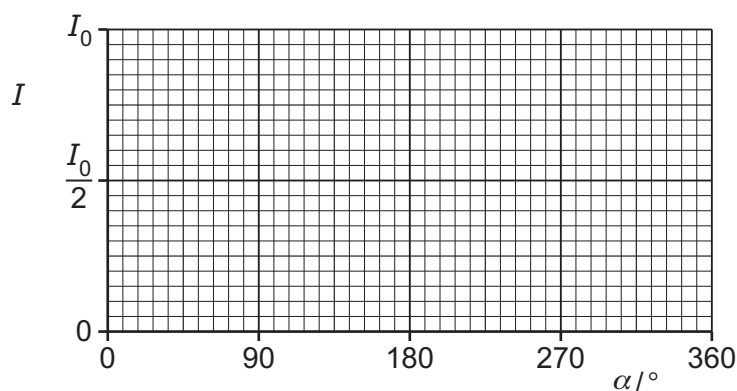
**Fig. 5.1**

The intensity of the light incident on the filter is I_0 .

The light is incident normally on the filter and the transmission axis of the filter is initially perpendicular to the plane of polarisation of the light.

The filter is now rotated through 360° about the direction of travel of the light wave.

- (i) On Fig. 5.2, sketch the variation of the intensity I of the transmitted light with the angle of rotation α as the filter is rotated through 360° from its initial position.

**Fig. 5.2**

[3]

- (ii) The amplitude of the incident light wave is A_0 when the intensity of the wave is I_0 .

Use Malus's law to determine, in terms of A_0 , the amplitude of the transmitted wave when $\alpha = 20^\circ$.

amplitude =

A_0 [4]

9. M/J 25/P23/Q6/b(i)

- (b) Light of wavelength 720 nm in a vacuum is incident normally on a diffraction grating as shown in Fig. 6.1.

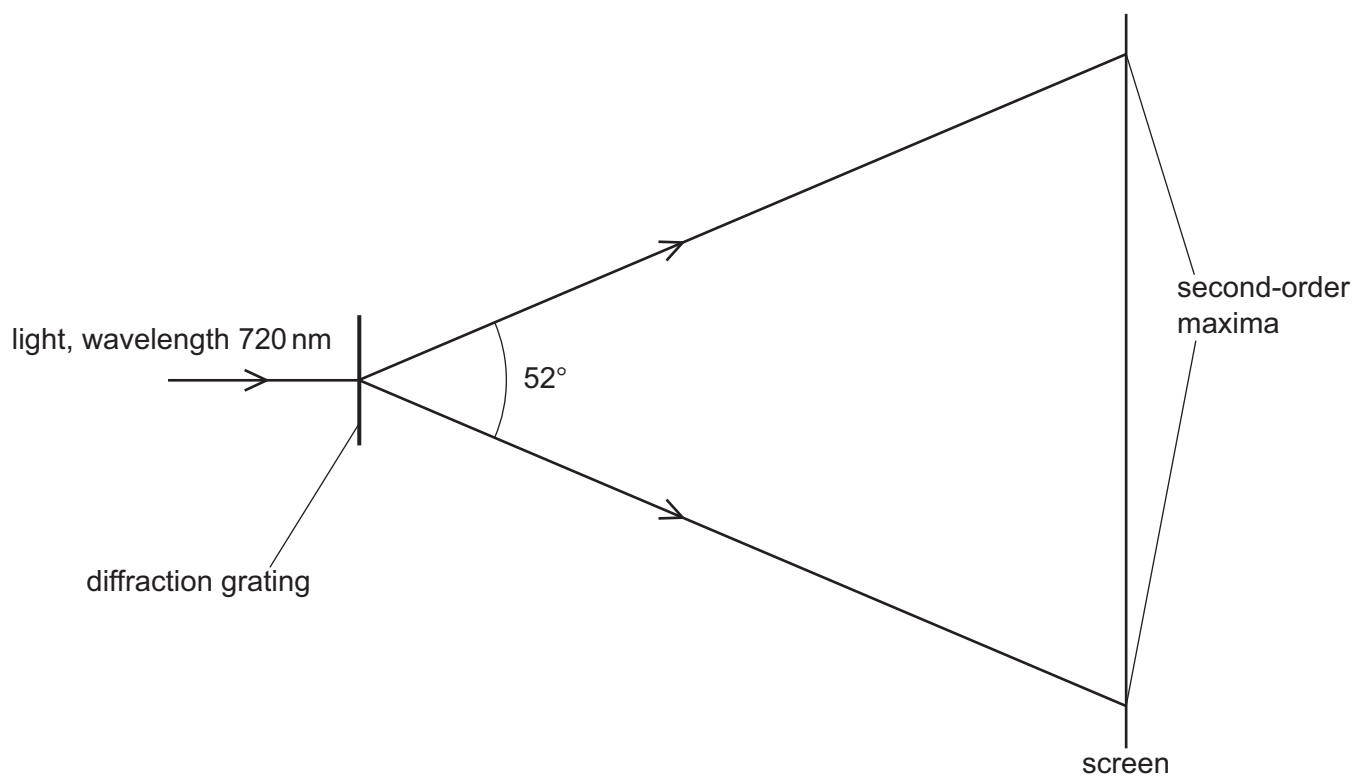


Fig. 6.1 (not to scale)

A screen is parallel to the grating. An interference pattern is seen on the screen and the angle between the **second**-order maxima is 52° .

- (i) Calculate the frequency of the light.

frequency =

Hz [2]

10. M/J 25/P24/Q4/a,b(i)

- 4 (a)** A source oscillates with frequency f to produce a progressive wave of wavelength λ . The source takes time t to produce n complete oscillations.

(i) State what is meant by a progressive wave.

[1]

(ii) State expressions, in terms of some or all of f , λ and n , for:

- the distance moved by a wavefront in time t

distance =

- time t .

time t =

[2]

(iii) Use your answers in (ii) to determine an expression for the speed v of the wave in terms of f and λ .

[1]

- (b)** Two identical microwave sources X and Y emit waves in phase. The sources are separated by a distance of 30 cm, as shown in Fig. 4.1.

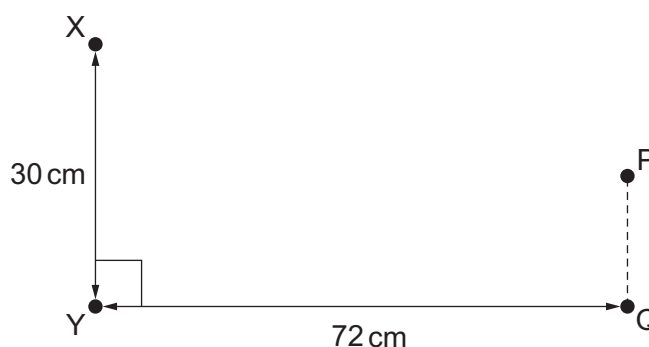


Fig. 4.1 (not to scale)

The intensity of the microwaves is to be investigated at points P and Q.

Line PQ is parallel to line XY. Distance XP is equal to distance YP. Distance YQ is 72 cm and angle XYQ is 90° .

The wavelength of the microwaves is 4.0 cm.

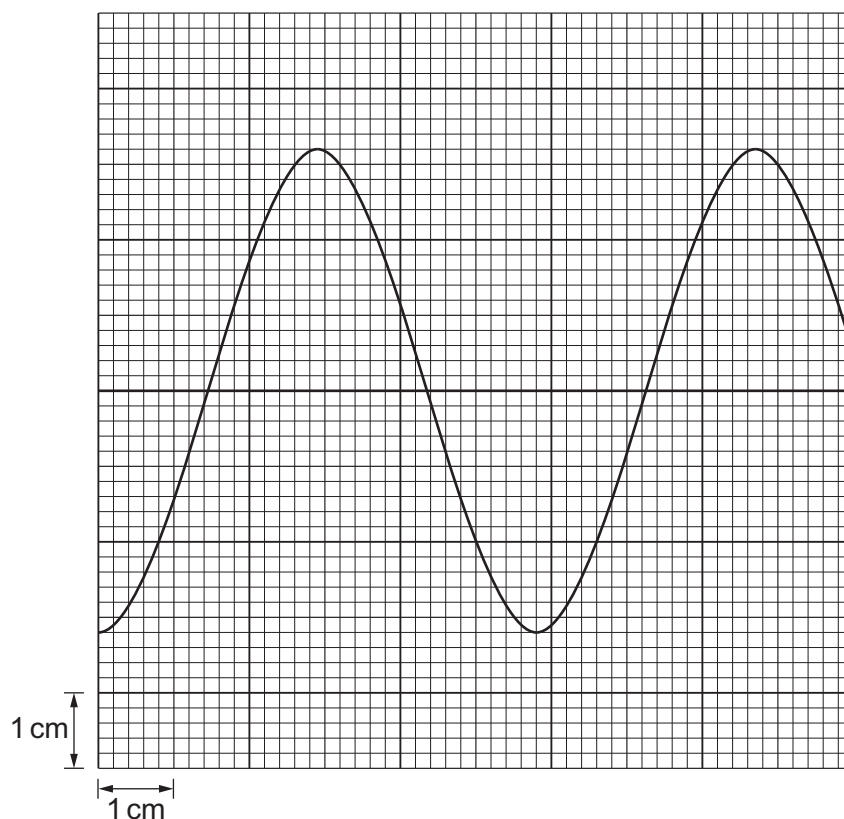
- (i) Calculate the frequency, in GHz, of the microwaves.

frequency =

GHz [2]

11. MAR 25/P22/Q5/a,b

- 5** A stationary loudspeaker emits sound of constant frequency. A microphone is placed near to the loudspeaker and connected to a cathode-ray oscilloscope (CRO). The trace on the screen of the CRO is shown in Fig. 5.1.

**Fig. 5.1**

The time-base of the CRO is set to $5.0 \times 10^{-4} \text{ s cm}^{-1}$.

- (a)** The speed of the sound emitted by the loudspeaker is 330 m s^{-1} .

Determine the wavelength of the sound.

wavelength = _____ m [3]

- (b)** The loudspeaker now moves in a straight line while emitting the same sound of constant frequency. The period of the trace on the CRO increases continuously.

Describe the motion of the loudspeaker.

[2]

12. O/N 24/P21/Q6/b(i)

- 6 Two coherent sources X and Y of microwaves of frequency 2.5×10^{10} Hz are a distance of 0.18 m apart in a vacuum, as shown in Fig. 6.1.

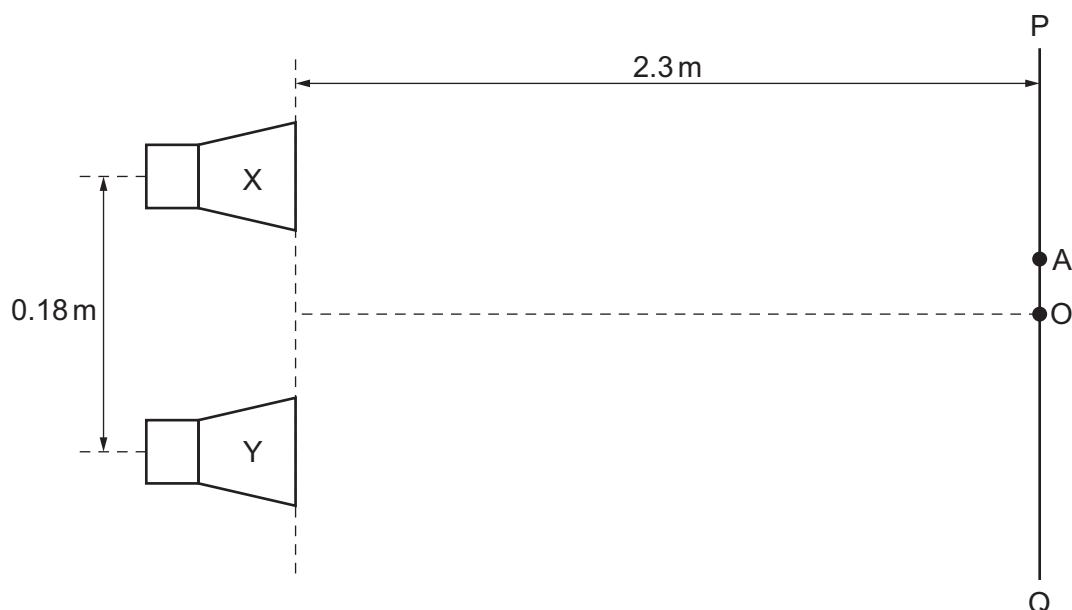


Fig. 6.1 (not to scale)

There is a phase difference of 90° between the waves emitted at the two sources.

A microwave detector moves along the line PQ, which is parallel to the line joining the two sources and 2.3 m away from it.

Point O is on the line PQ at a position that is equidistant from the two sources.

Point A is the position on line PQ where the intensity of the microwaves is the greatest.

- (b) (i) Show that the wavelength of the microwaves is 0.012 m.

[2]

13. O/N 24/P22/Q4/a

- 4 (a) With reference to the direction of transfer of energy, compare the oscillations of transverse and longitudinal progressive waves.

[2]

14. O/N 24/P23/Q5/a(iii),b

- 5 (a) A stationary wave is formed on a string XY that has a length of 0.48 m. Fig. 5.1 shows the string at one instant in time.

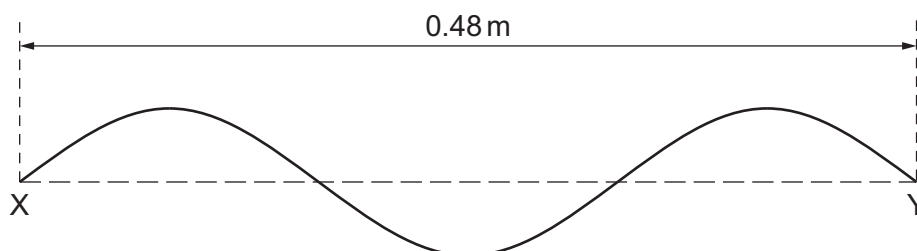


Fig. 5.1

The speed of the wave on the string is 1400 m s^{-1} .

- (iii) Calculate the frequency of the wave.

frequency = _____ Hz [2]

- (b) A source of sound waves of frequency 780 Hz is on a rotating platform. The speed of the source is 39 m s^{-1} .

The sound is detected by an observer that is a large distance from the rotating platform, as shown in Fig. 5.2.

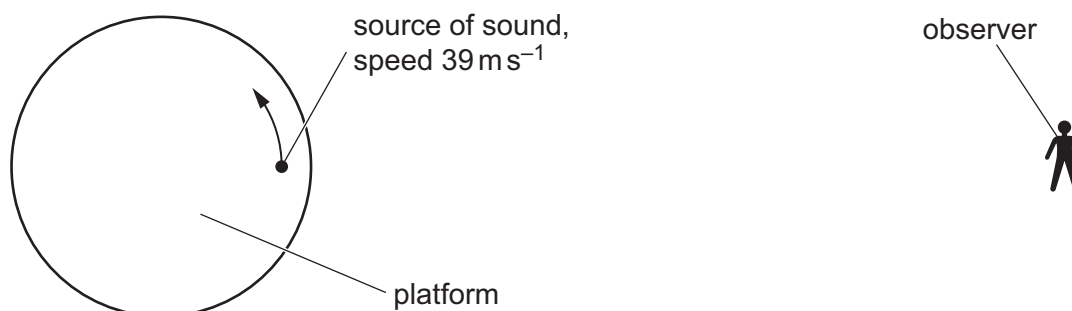


Fig. 5.2 (not to scale)

- (i) The speed of sound in air is 320 m s^{-1} .

Calculate the maximum frequency of the sound detected by the observer.

maximum frequency = _____ Hz [2]

- (ii) At time $t = 0$, the observer detects the sound emitted by the source when it was in the position shown in Fig. 5.2.

On Fig. 5.3, sketch the variation with t of the frequency f of the sound detected by the observer for one complete rotation of the platform. Calculations are not required.

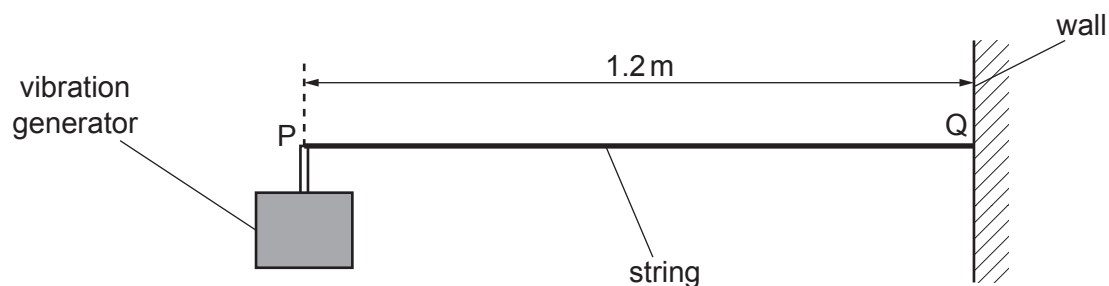


Fig. 5.3

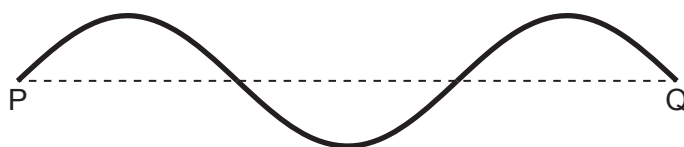
[2]

15. M/J 24/P21/Q5/e

- 5** A stretched string PQ has length 1.2 m. One end of the string is attached to a vibration generator and the other end is attached to a wall, as shown in Fig. 5.1.

**Fig. 5.1**

The vibration generator is switched on and a stationary wave is formed on the string. The string is shown at one instant of time in Fig. 5.2.

**Fig. 5.2**

- (e)** Calculate the speed of the progressive waves on the stretched string.

speed =

ms^{-1} [2]

16. M/J 24/P22/Q6/b(i)

- 6** Light of a single frequency is incident normally on a diffraction grating. An interference pattern of bright and dark fringes forms on the semicircular screen shown in Fig. 6.1.

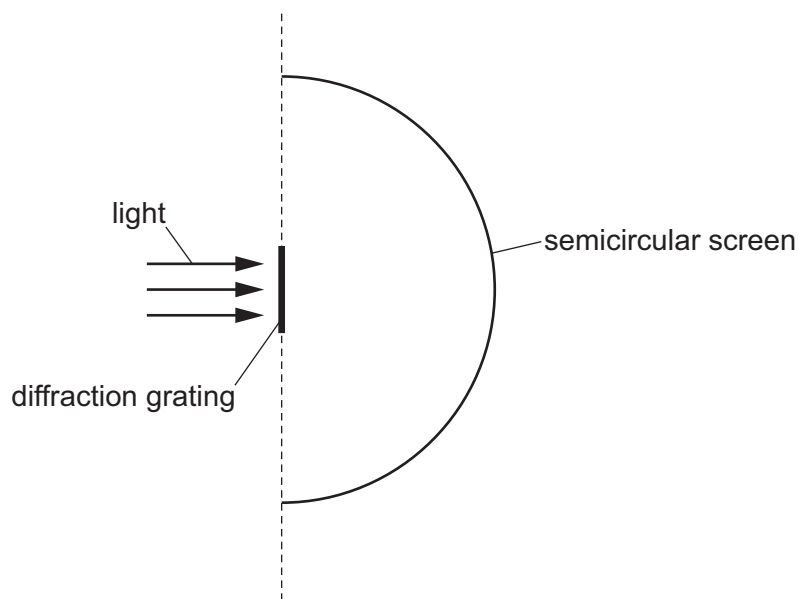


Fig. 6.1 (not to scale)

The light has wavelength 520 nm.

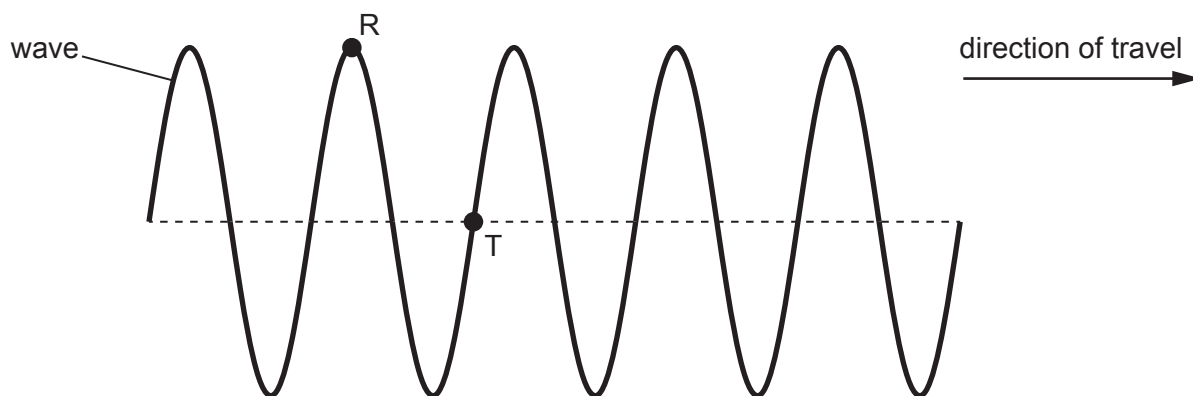
The separation of the lines in the grating is 3.8×10^{-6} m.

- (b)** The light is replaced with red light of a single frequency.
- (i)** State whether the frequency of the red light is greater than, less than or the same as the frequency of the original light.

[1]

17. M/J 24/P23/Q4/a,b,c

- 4 A progressive transverse wave travelling from left to right is shown at an instant in time in Fig. 4.1.

**Fig. 4.1**

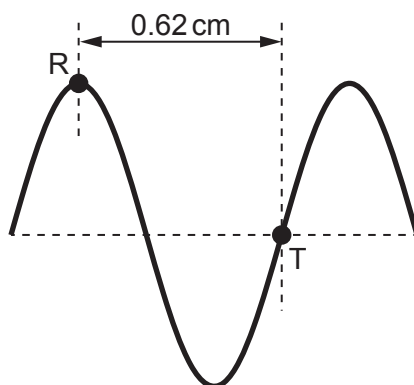
R and T are points on the wave.

- (a) State the phase difference between the points R and T.

phase difference = ° [1]

- (b) On Fig. 4.1, draw an arrow at point T to show the direction of movement of point T at the instant shown. [1]

- (c) The horizontal distance between R and T is 0.62 cm, as shown in Fig. 4.2.

**Fig. 4.2** (not to scale)

The speed of the wave is 0.27 m s^{-1} .

Calculate the frequency of the wave.

frequency = Hz [3]

18. MAR 24/P22/Q1/c

- (c) A light meter is used to measure the intensity of light in a classroom. Daylight is incident normally on the sensor of the meter. The sensor has an area of 2.2 cm^2 . The reading on the meter is 950 W m^{-2} .

Calculate the power of the daylight incident on the sensor.

power = W [3]

19. MAR 24/P22/Q5/a,b,c

- 5 (a)** By reference to the direction of propagation of energy, state what is meant by a transverse wave.

[1]

- (b)** A space telescope is designed to detect electromagnetic radiation with wavelengths in the range $12\text{ }\mu\text{m}$ to $28\text{ }\mu\text{m}$.

State the region of the electromagnetic spectrum for this radiation.

[1]

- (c)** A detector on another space telescope detects an electromagnetic wave. The signal from the detector is transmitted to Earth and displayed on an oscilloscope as shown in Fig. 5.1. The frequency of the signal displayed on the oscilloscope is equal to the frequency of the detected electromagnetic wave.

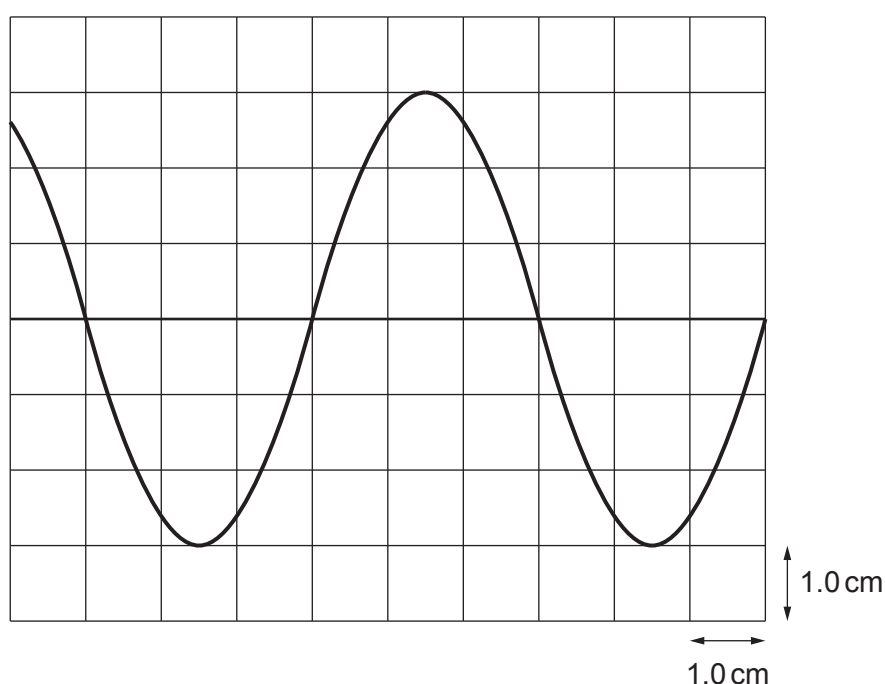


Fig. 5.1

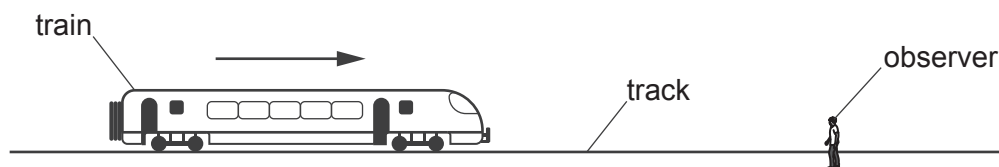
The time-base setting on the oscilloscope is $5.0 \times 10^{-15} \text{ s cm}^{-1}$.

Calculate the wavelength of the detected electromagnetic wave.

wavelength = m [3]

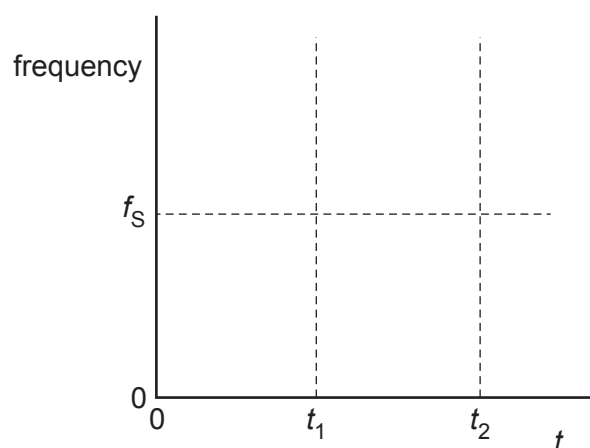
20. O/N 23/P21/Q5/a,b

- 5** A train travels at a constant high speed along a straight horizontal track towards an observer standing adjacent to the track, as shown in Fig. 5.1.

**Fig. 5.1**

The train sounds its horn continuously as it approaches the observer, from time $t = 0$ until it is well past the observer at time $t = t_2$. The train passes the observer at time $t = t_1$. The horn emits a sound wave of constant frequency f_s .

- (a)** On Fig. 5.2, sketch the variation of the frequency of sound heard by the observer with time t , from time $t = 0$ to $t = t_2$.

**Fig. 5.2**

[1]

- (b)** At a particular time, the sound waves at the observer have an intensity of $4.7 \times 10^{-3} \text{ W m}^{-2}$. The waves at the observer are incident at right angles on a circular detector of radius 2.8 cm.

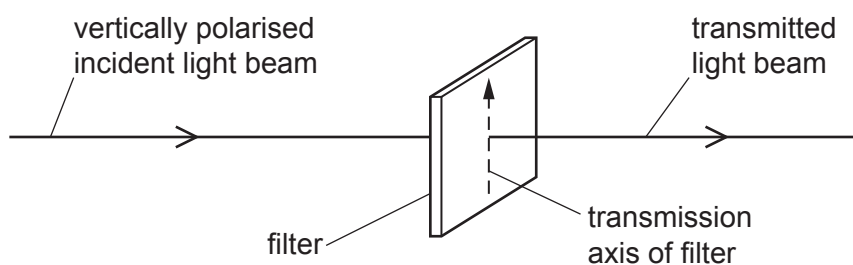
Calculate the power P of the waves incident on the detector.

$P =$

W [3]

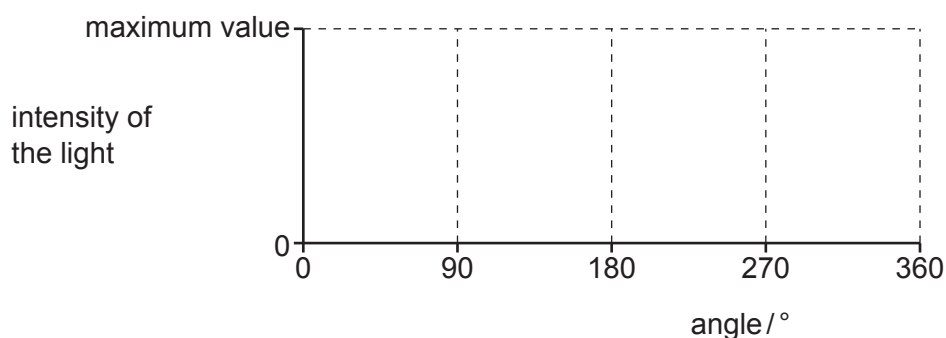
21. O/N 23/P22/Q5/a

- 5 (a)** A beam of vertically polarised light is incident normally on a polarising filter, as shown in Fig. 5.1.

**Fig. 5.1**

- (i) The transmission axis of the filter is initially vertical. The filter is then rotated through an angle of 360° while the plane of the filter remains perpendicular to the beam.

On Fig. 5.2, sketch a graph to show the variation of the intensity of the light in the transmitted beam with the angle through which the transmission axis is rotated.

**Fig. 5.2**

[2]

- (ii) The intensity of the light in the incident beam is 7.6 W m^{-2} . When the transmission axis of the filter is at angle θ to the vertical, the light intensity of the transmitted beam is 4.2 W m^{-2} .

Calculate angle θ .

$\theta =$ ° [2]

22. O/N 23/P23/Q5/b(ii)

- 5 Two point sources, A and B, produce coherent electromagnetic waves. The waves from A and B are emitted in phase and have wavelength λ , as shown in Fig. 5.1.

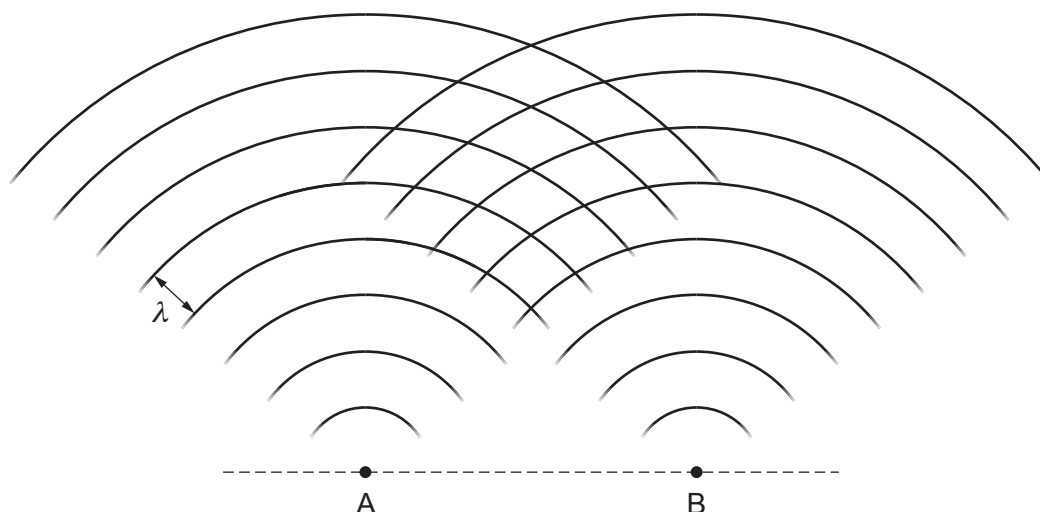


Fig. 5.1 (not to scale)

The lines on Fig. 5.1 represent wavefronts. All the points on a wavefront are in phase.

- (b) The waves in air have a wavelength of 2.9×10^{-5} m.

An interference pattern is detected along a line parallel to AB and at a perpendicular distance of 140 m from AB. The spacing between adjacent interference maxima is 1.2 cm.

- (ii) State the principal region of the electromagnetic spectrum to which the waves belong.

[1]

23. O/N 23/P23/Q6/a,b

- 6 A train travels at constant speed along a straight horizontal track towards an observer standing adjacent to the track, as shown in Fig. 6.1.

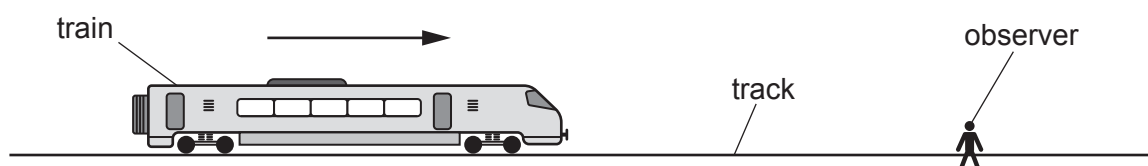


Fig. 6.1

The train sounds its horn continuously as it approaches the observer. The horn emits a sound of constant frequency 251 Hz. The frequency of sound heard by the observer is 291 Hz. The speed of sound in air is 340 ms^{-1} .

- (a) Calculate the speed of the train.

speed = ms^{-1} [2]

- (b) The train approaches and then passes the observer. The intensity I of the sound heard by the observer varies with the distance d of the horn from the observer.

When the horn is at a distance x_0 from the observer, the intensity I of the sound heard is I_0 and the amplitude A of the sound wave at the observer is A_0 .

Fig. 6.2 shows the variation with d/x_0 of I/I_0 as the train moves away from the observer.

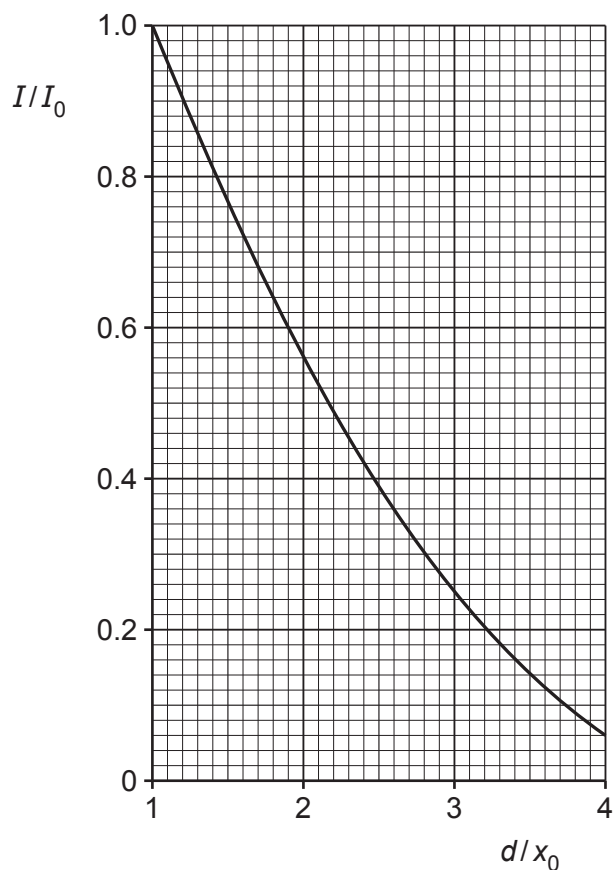


Fig. 6.2

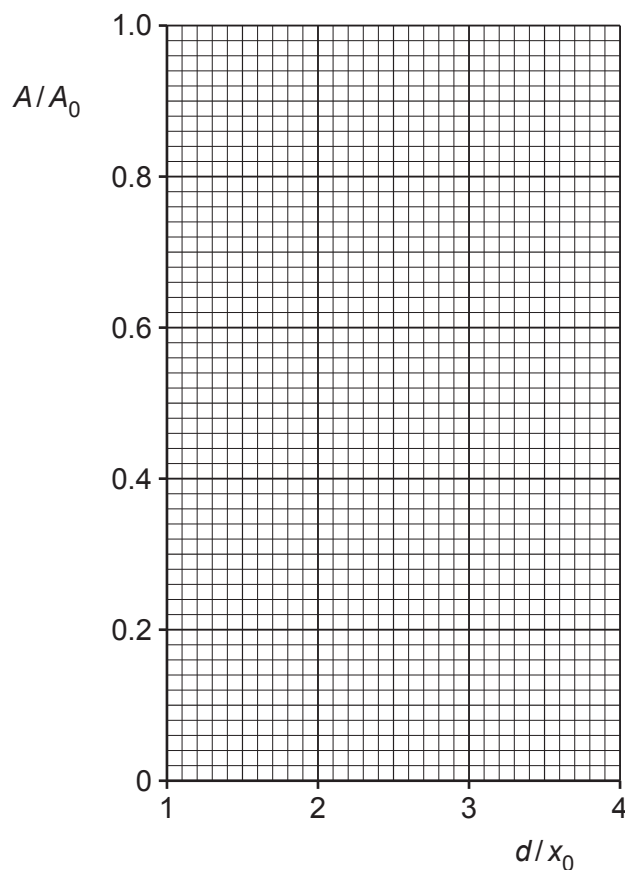


Fig. 6.3

- (i) State the relationship between amplitude A and intensity I for a progressive wave.

[1]

- (ii) On Fig. 6.3, sketch the variation with d/x_0 of A/A_0 .

[2]

24. M/J 23/P22/Q5

booklet (scan + OCR)

- (a)** A progressive wave travels through a medium. The wave causes a particle of the medium to vibrate along a line P. The energy of the wave propagates along a line Q.

Compare the directions of lines P and Q if the wave is:

- (i)** a transverse wave

..... [1]

- (ii)** a longitudinal wave.

..... [1]

- (b)** A tube is closed at one end. A loudspeaker is placed near the other end of the tube, as shown in Fig. 5.1.

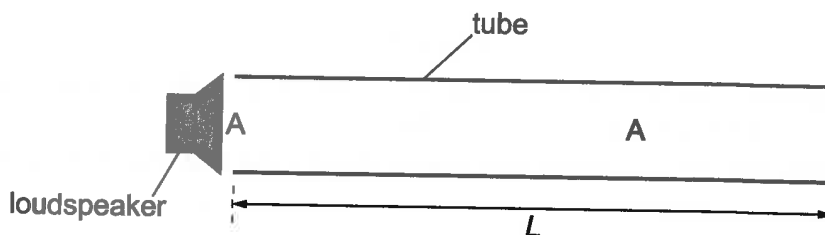


Fig. 5.1 (not to scale)

The loudspeaker emits sound of frequency 1.7 kHz . The speed of sound in the air in the tube is 340 m s^{-1} . A stationary wave is formed with an antinode A at the open end of the tube. There is only one other antinode A inside the tube, as shown in Fig. 5.1.

Determine:

- (i)** the wavelength of the sound

wavelength = m [2]

- (ii)** the length L of the tube

$L = \dots\dots\dots \text{m}$ [1]

- (iii)** the maximum wavelength of the sound from the loudspeaker that can produce a stationary wave in the tube.

maximum wavelength = m [1]

- (c) Two polarising filters are arranged so that their planes are vertical and parallel. The first filter has its transmission axis at an angle of 35° to the vertical and the second filter has its transmission axis at angle α to the vertical, as shown in Fig. 5.2.

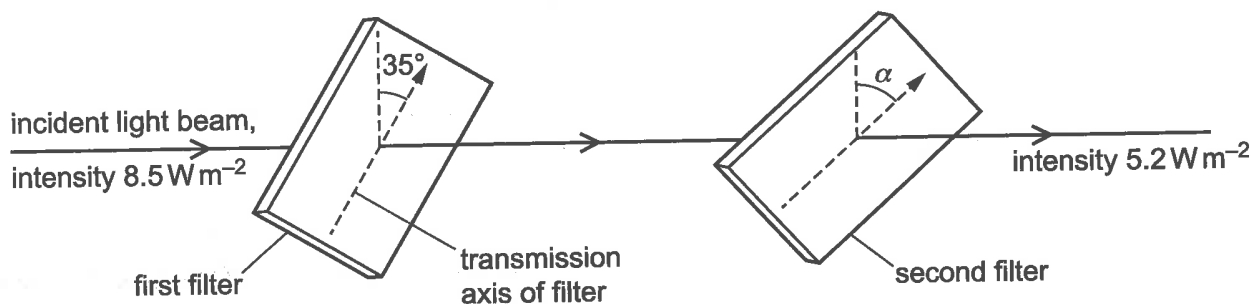


Fig. 5.2

Angle α is greater than 35° and less than 90° . A beam of vertically polarised light of intensity 8.5 W m^{-2} is incident normally on the first filter.

- (i) Show that the intensity of the light transmitted by the first filter is 5.7 W m^{-2} .

[1]

- (ii) The intensity of the light transmitted by the second filter is 5.2 W m^{-2} . Calculate angle α .

$\alpha = \dots\dots\dots^\circ$ [2]

25. M/J 23/P23/Q4

booklet (scan + OCR)

- (a) For a progressive wave, state what is meant by the frequency.

..... [1]

- (b) A loudspeaker, microphone and cathode-ray oscilloscope (CRO) are arranged as shown in Fig. 4.1.

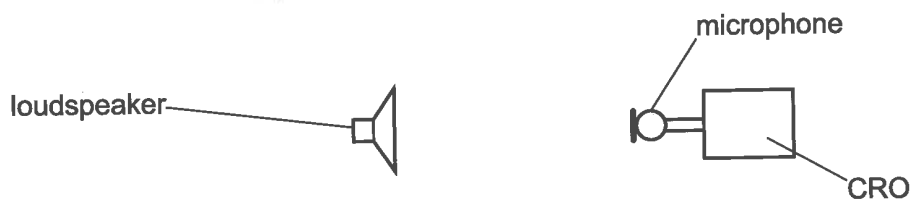


Fig. 4.1

The loudspeaker is emitting a sound wave which is detected by the microphone and displayed on the screen of the CRO as shown in Fig. 4.2.

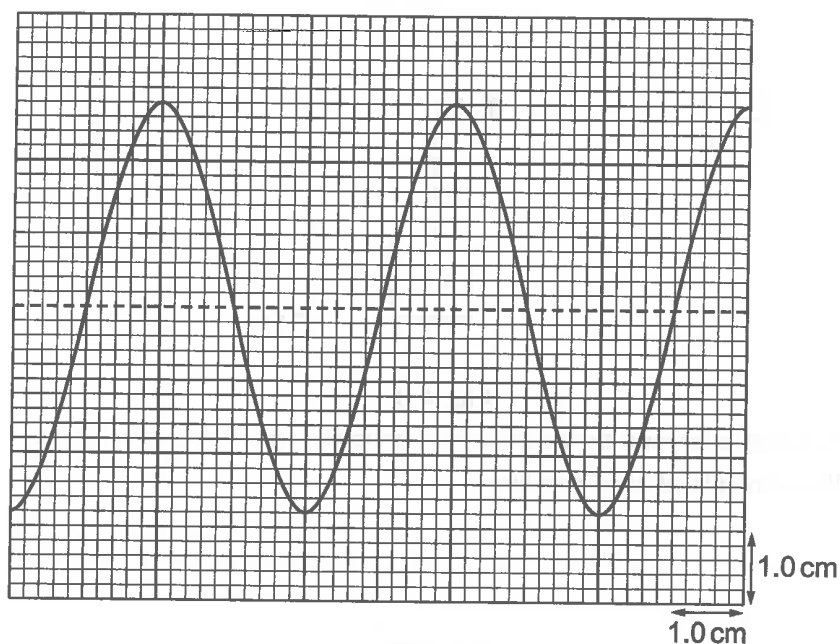


Fig. 4.2

The time-base on the CRO is set to 0.50 ms cm^{-1} and the y-gain is set to 0.20 V cm^{-1} . Calculate:

- (i) the frequency of the sound wave

frequency = Hz [2]

- (ii) the amplitude of the signal received by the CRO.

amplitude = V [1]

- (c) The intensity of the sound wave in (b) is reduced to a quarter of its original intensity without a change in frequency. Assume that the amplitude of the signal received by the CRO is proportional to the amplitude of the sound wave.

On Fig. 4.2, sketch the trace that is now seen on the screen of the CRO. [3]

(d) A metal sheet is now placed in front of the loudspeaker in (b), as shown in Fig. 4.3.

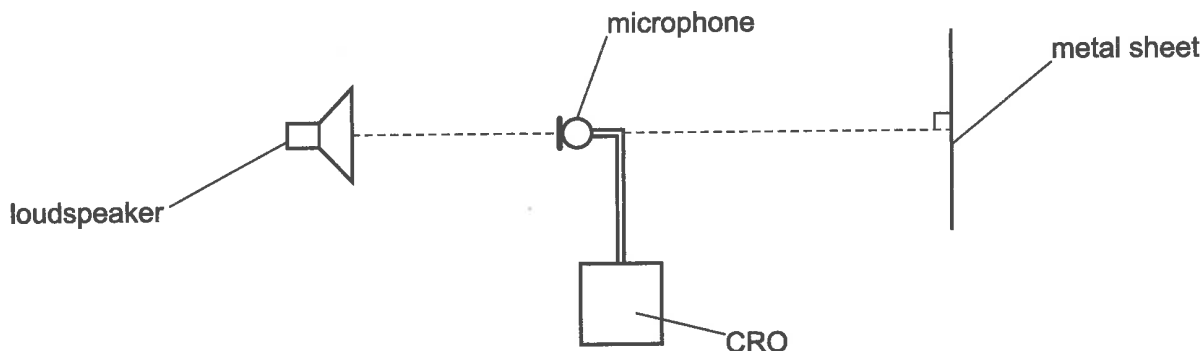


Fig. 4.3

A stationary wave is formed between the loudspeaker and the metal sheet.

(i) State the principle of superposition.

.....
 [2]

(ii) The initial position of the microphone is such that the trace on the CRO has an amplitude minimum. It is now moved a distance of 1.05 m away from the loudspeaker along the line joining the loudspeaker and metal sheet.

As the microphone moves, it passes through three positions where the trace has an amplitude maximum before ending at a position where the trace has an amplitude minimum.

Determine the wavelength of the sound wave.

wavelength = m [2]

(iii) Use your answers in (b)(i) and (d)(ii) to determine the speed of the sound in the air.

speed = ms^{-1} [2]

26. O/N 22/P22/Q5

booklet (scan + OCR)

(a) Two progressive sound waves meet to form a stationary wave. The two waves have the same amplitude, wavelength, frequency and speed.

State the other condition that must be fulfilled by the two waves in order for them to produce the stationary wave.

..... [1]

- (b) A stationary wave is formed on a string that is stretched between two fixed points A and B. Fig. 5.1 shows the string at time $t = 0$ when each point is at its maximum displacement.

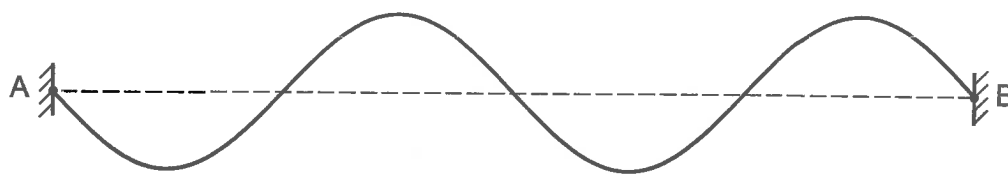


Fig. 5.1

Distance AB is 0.80 m. The period of the stationary wave is 0.016 s.

- (i) On Fig. 5.1, sketch a solid line to show the position of the string:

- at time $t = 0.004$ s (label this line P)
- at time $t = 0.024$ s (label this line Q).

[2]

- (ii) Determine the speed of a progressive wave along the string.

speed = ms^{-1} [3]

- (c) A beam of vertically polarised light of intensity I_0 is incident normally on a polarising filter that has its transmission axis at 30° to the vertical, as shown in Fig. 5.2.

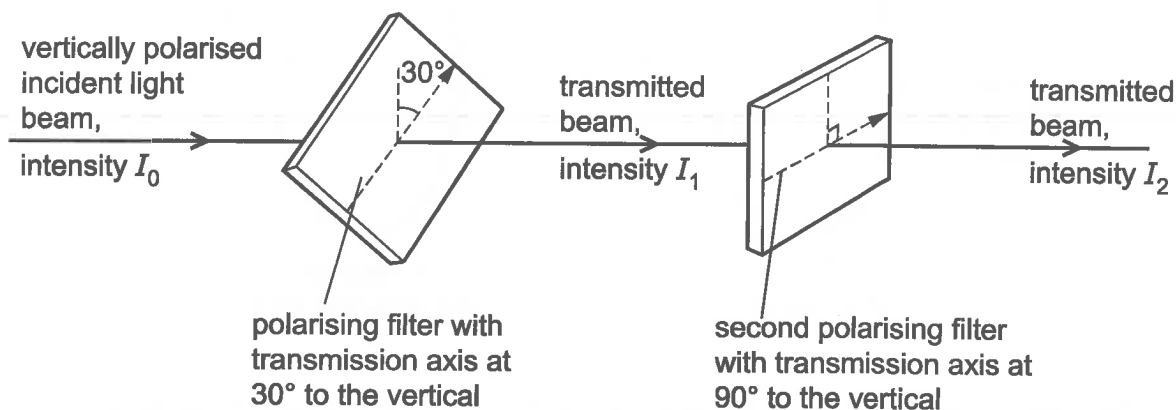


Fig. 5.2

The transmitted light from the first polarising filter has intensity I_1 . This light is then incident normally on a second polarising filter that has its transmission axis at 90° to the vertical. The transmitted light from the second filter has intensity I_2 .

Calculate:

- (i) the ratio $\frac{I_1}{I_0}$

$\frac{I_1}{I_0} = \dots\dots\dots$ [2]

(ii) the ratio $\frac{I_2}{I_0}$.

$$\frac{I_2}{I_0} = \dots\dots\dots [2]$$

27. O/N 22/P21/Q4

booklet (scan + OCR)

(a) Polarisation is a phenomenon associated with light waves but not with sound waves.

(i) State the meaning of polarisation.

..... [1]

(ii) State why light waves can be plane polarised but sound waves cannot.

..... [1]

(b) Two polarising filters A and B are positioned so that their planes are parallel to each other and perpendicular to a central axis line XY, as shown in Fig. 4.1.

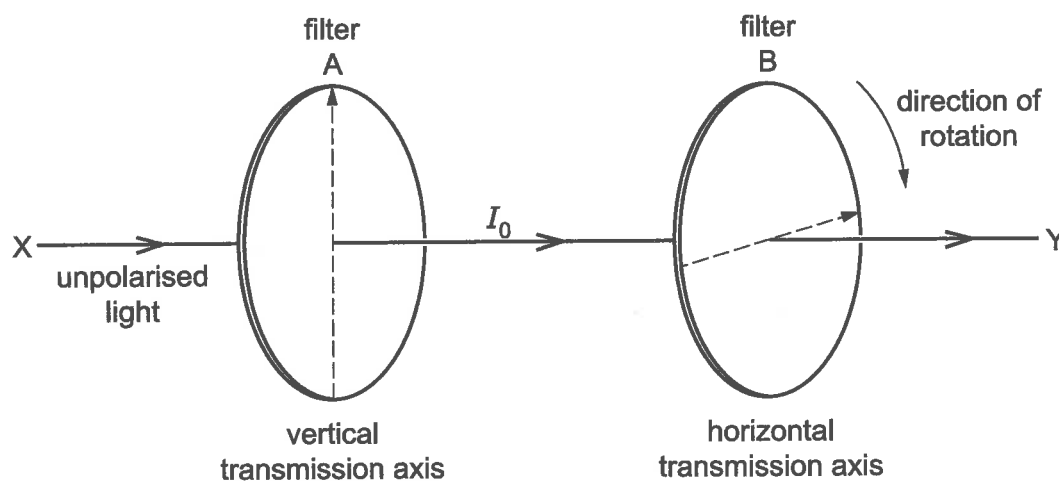


Fig. 4.1

The transmission axis of filter A is vertical and the transmission axis of filter B is horizontal. Unpolarised light of a single frequency is directed along the line XY from a source positioned at X. The light emerging from filter A is vertically plane polarised and has intensity I_0 . Filter B is rotated from its starting position about the line XY, as shown in Fig. 4.1.

After rotation, the intensity of the light emerging from filter B is $\frac{1}{4} I_0$.

Calculate the angle of rotation of filter B from its starting position.

angle of rotation = ° [3]

- (c) A microwave of intensity I_0 and amplitude A_0 meets another microwave of the same frequency and of intensity $\frac{1}{4}I_0$ travelling in the opposite direction. Both microwaves are vertically plane polarised and superpose where they meet.

- (i) Explain, without calculation, why these two waves cannot form a stationary wave with zero amplitude at its nodes.

.....

..... [2]

- (ii) Determine, in terms of A_0 , the maximum amplitude of the wave formed.

maximum amplitude = A_0 [3]

28. O/N 22/P23/Q4

booklet (scan + OCR)

- (a) A progressive longitudinal wave travels through a medium from left to right. Fig. 4.1 shows the positions of some of the particles of the medium at time t_0 and a graph showing the particle displacements at the same time t_0 .

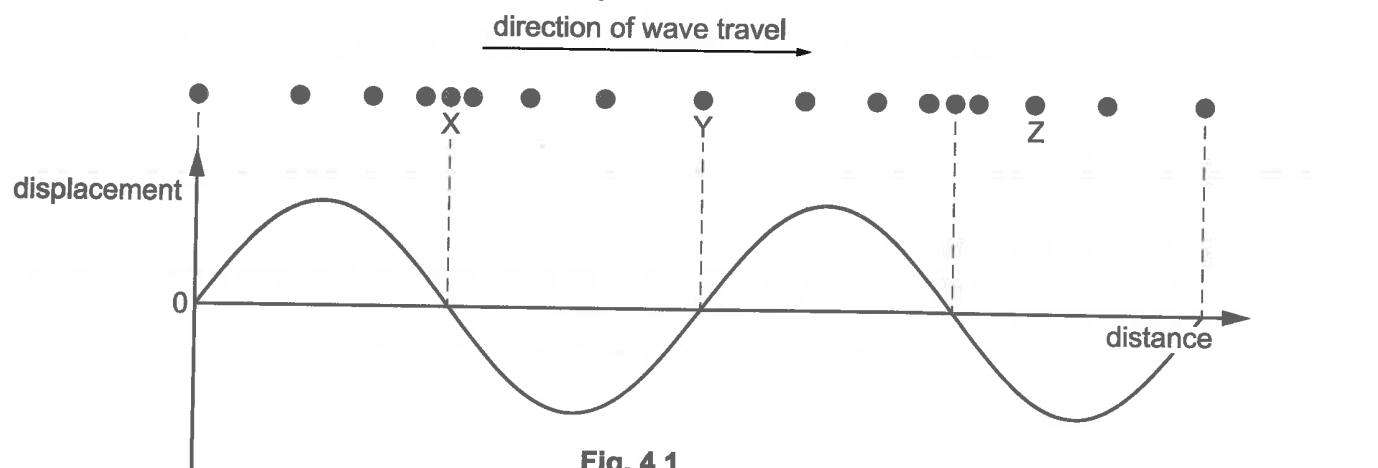


Fig. 4.1

Particle displacements to the right of their equilibrium positions are shown as positive on the graph and particle displacements to the left are shown as negative on the graph.

The period of the wave is T .

- (i) On Fig. 4.1, draw circles around two particles which are exactly one wavelength apart. [1]
- (ii) On Fig. 4.1, sketch a line on the graph to represent the displacements of the particles for the longitudinal wave at time $t_0 + \frac{T}{4}$. [3]
- (iii) State the direction of motion of particle Z at time $t_0 + \frac{T}{4}$.

..... [1]

- (b) The frequency of the wave in (a) is 16 kHz. The distance between particles X and Y is 0.19 m. Calculate the speed of the wave as it travels through the medium.

speed = ms^{-1} [3]

- (c) A longitudinal sound wave is travelling through a solid. The initial intensity of the wave is I_0 . The frequency of the wave remains constant and the amplitude falls to half of its original value. Determine, in terms of I_0 , the final intensity of the wave.

intensity = I_0 [2]

- (d) The sound wave in (c) now meets another sound wave travelling in the opposite direction.

- (i) State a condition necessary for these two waves to form a stationary wave.

..... [1]

- (ii) State **two** ways in which a stationary wave differs from a progressive wave.

1 [1]

2 [2]

29. M/J 22/P21/Q5

booklet (scan + OCR)

A horizontal string is stretched between two fixed points A and B. A vibrator is used to oscillate the string and produce an observable stationary wave.

At one instant, the moving string is straight, as shown in Fig. 5.1.

**Fig. 5.1**

The dots in the diagram represent the positions of the nodes on the string. Point P on the string is moving downwards. The wave on the string has a speed of 35 m s^{-1} and a period of 0.040 s .

- (a) Explain how the stationary wave is formed on the string.

.....
 [2]

- (b) On Fig. 5.1, sketch a line to show a possible position of the string a quarter of a cycle later than the position shown in the diagram. [1]

- (c) Determine the horizontal distance from A to B.

distance = m [3]

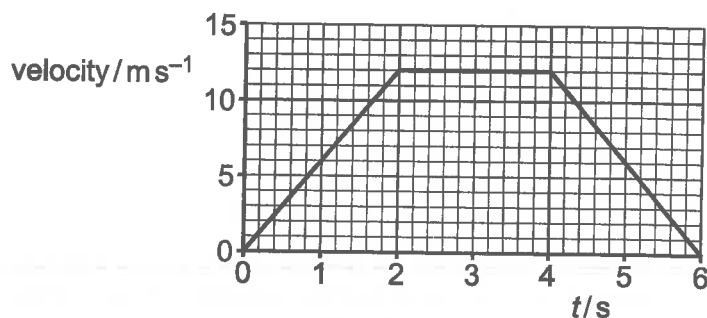
- (d) A particle on the string has zero displacement at time $t = 0$. From time $t = 0$ to time $t = 0.060 \text{ s}$, the particle moves through a total distance of 72 mm . Calculate the amplitude of oscillation of the particle.

amplitude = mm [2]

30. O/N 21/P22/Q4

booklet (scan + OCR)

A child sits on the ground next to a remote-controlled toy car. At time $t = 0$, the car begins to move in a straight line directly away from the child. The variation with time t of the velocity of the car along this line is shown in Fig. 4.1.

**Fig. 4.1**

The car's horn continually emits sound of frequency 925 Hz between time $t = 0$ and time $t = 6.0$ s. The speed of the sound in the air is 338 m s^{-1} .

- (a) Describe qualitatively the variation, if any, in the frequency of the sound heard, by the child, that was emitted from the car horn:

(i) from time $t = 0$ to time $t = 2.0$ s

..... [1]

(ii) from time $t = 4.0$ s to time $t = 6.0$ s.

..... [1]

- (b) Determine the frequency, to three significant figures, of the sound heard, by the child, that was emitted from the car horn at time $t = 3.0$ s.

frequency = Hz [2]

- (c) Determine the time taken for the sound emitted at time $t = 4.0$ s to travel to the child.

time taken = s [2]

31. O/N 21/P21/Q4

booklet (scan + OCR)

- (a) By reference to the direction of transfer of energy, state what is meant by a *longitudinal wave*.

..... [1]

- (b) A vehicle travels at constant speed around a wide circular track. It continuously sounds its horn, which emits a single note of frequency 1.2 kHz. An observer is a large distance away from the track, as shown in the view from above in Fig. 4.1.

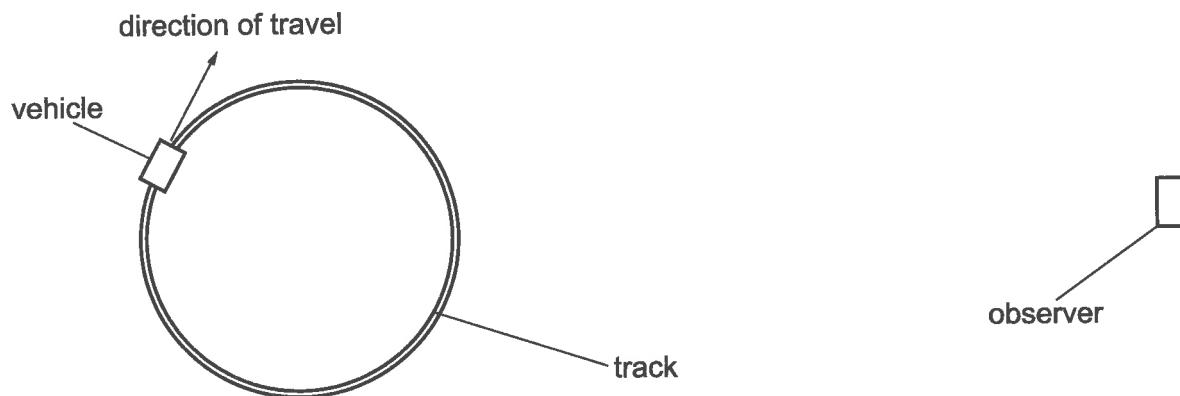


Fig. 4.1 (not to scale)

Fig. 4.2 shows the variation with time of the frequency f of the sound of the horn that is detected by the observer. The time taken for the vehicle to travel once around the track is T .

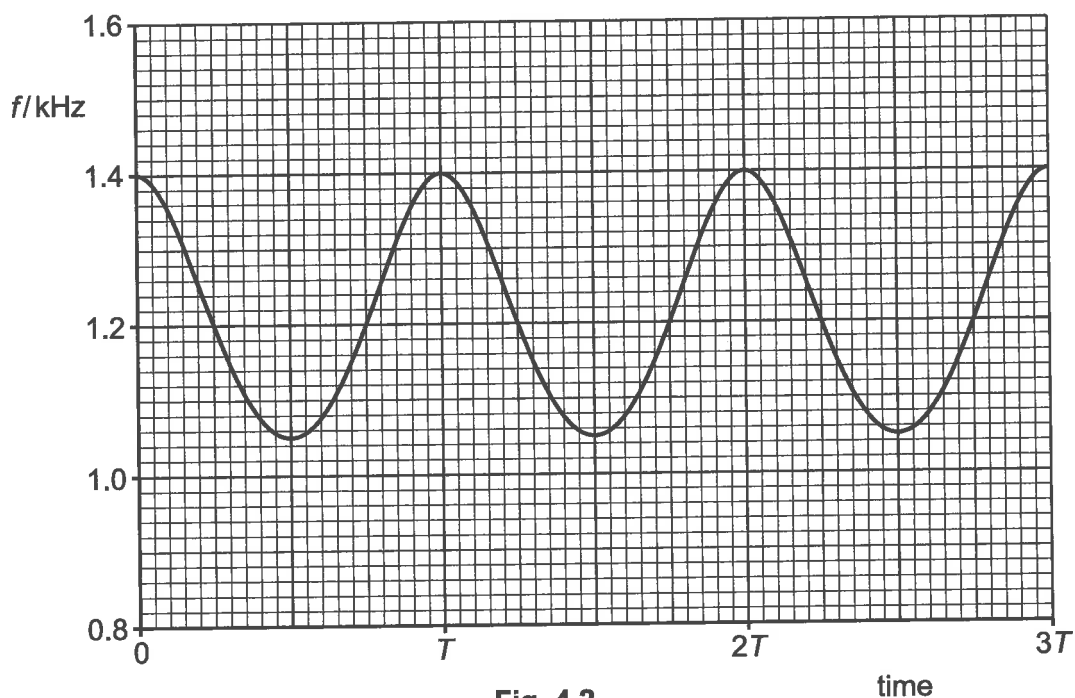


Fig. 4.2

- (i) Explain why the frequency of the sound detected by the observer is sometimes above and sometimes below 1.2 kHz.

.....

..... [2]

- (ii) State the name of the phenomenon in (b)(i).

..... [1]

- (iii) On Fig. 4.1, mark with a letter X the position of the vehicle when it emitted the sound that is detected at time T . [1]

- (iv) On Fig. 4.1, mark with a letter Y the position of the vehicle when it emitted the sound that is detected at time $\frac{9T}{4}$. [1]

- (c) The speed of the sound in the air is 320 m s^{-1} .

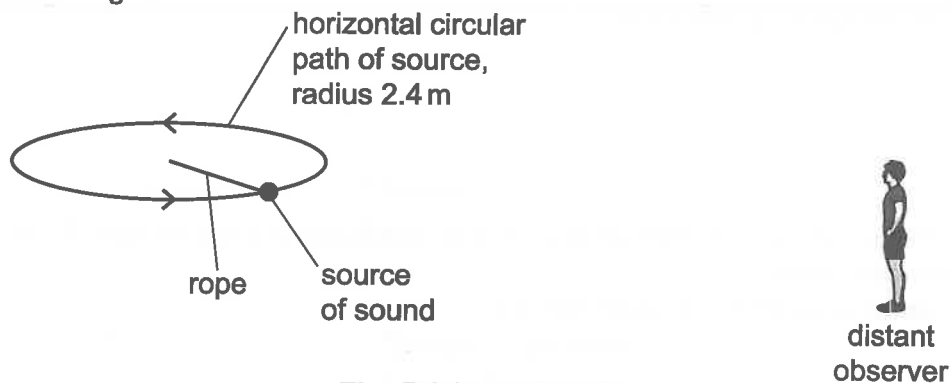
Use Fig. 4.2 to determine the speed of the vehicle in (b).

speed = m s^{-1} [3]

32. MAR 21/P22/Q5

booklet (scan + OCR)

A source of sound is attached to a rope and then swung at a constant speed in a horizontal circle, as illustrated in Fig. 5.1.

**Fig. 5.1** (not to scale)

The source moves with a speed of 12.0 m s^{-1} and emits sound of frequency 951 Hz . The speed of the sound in the air is 330 m s^{-1} . An observer, standing a very long distance away from the source, hears the sound.

- (a) Calculate the minimum frequency, to three significant figures, of the sound heard by the observer.

minimum frequency = Hz [2]

- (b) The circular path of the source has a radius of 2.4 m .

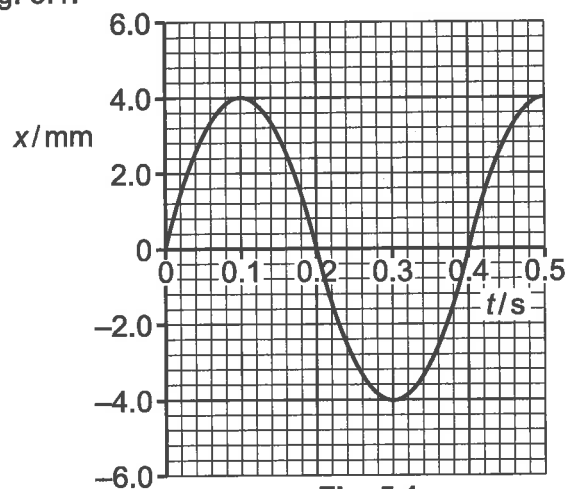
Determine the shortest time interval between the observer hearing sound of minimum frequency and the observer hearing sound of maximum frequency.

time interval = s [2]

33. O/N 20/P21/Q5

booklet (scan + OCR)

A progressive wave Y passes a point P. The variation with time t of the displacement x for the wave at P is shown in Fig. 5.1.

**Fig. 5.1**

The wave has a wavelength of 8.0 cm.

(a) Determine the speed of the wave.

speed = ms^{-1} [2]

(b) A second wave Z has wavelength 8.0 cm and amplitude 2.0 mm at point P. Waves Y and Z have the same speed.

For the waves at point P, calculate the ratio

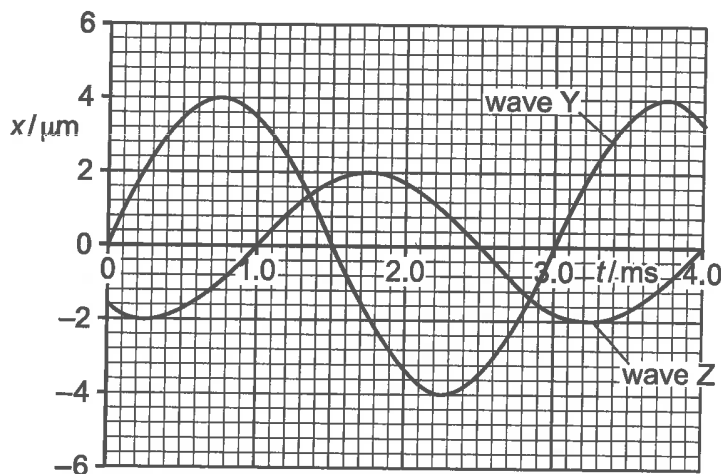
$$\frac{\text{intensity of wave Z}}{\text{intensity of wave Y}}.$$

ratio = [3]

34. M/J 20/P23/Q4

booklet (scan + OCR)

Two progressive sound waves Y and Z meet at a fixed point P. The variation with time t of the displacement x of each wave at point P is shown in Fig. 4.1.

**Fig. 4.1**

(a) Use Fig. 4.1 to state **one** quantity of waves Y and Z that is:

(i) the same

..... [1]

(ii) different.

..... [1]

(b) State and explain whether waves Y and Z are coherent.

..... [1]

(c) Determine the phase difference between the waves.

phase difference =° [1]

(d) The two waves superpose at P. Use Fig. 4.1 to determine the resultant displacement at time $t = 0.75$ ms.

resultant displacement = μm [1]

(e) The intensity of wave Y at point P is I .

Determine, in terms of I , the intensity of wave Z.

intensity = [2]

(f) The speed of wave Z is 330 ms^{-1} .

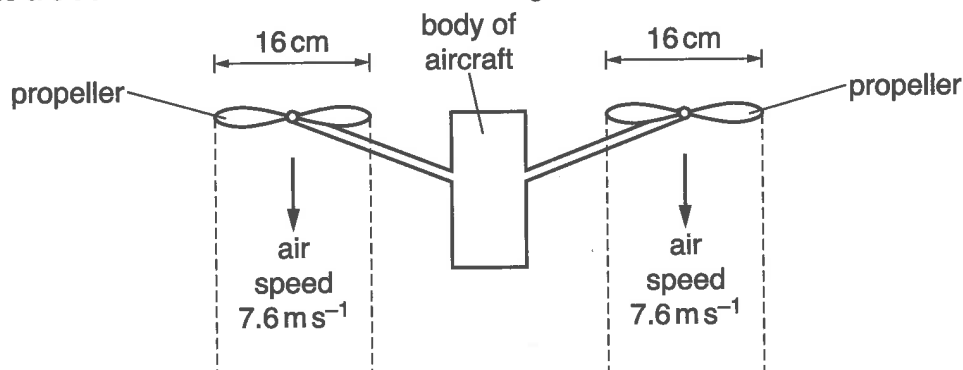
Determine the wavelength of wave Z.

wavelength = m [3]

35. O/N 19/P21/Q3

booklet (scan + OCR)

A small remote-controlled model aircraft has two propellers, each of diameter 16 cm. Fig. 3.1 is a side view of the aircraft when hovering.

**Fig. 3.1**

Air is propelled vertically downwards by each propeller so that the aircraft hovers at a fixed position. The density of the air is 1.2 kg m^{-3} . Assume that the air from each propeller moves with a constant speed of 7.6 m s^{-1} in a uniform cylinder of diameter 16 cm. Also assume that the air above each propeller is stationary.

- (a) Show that, in a time interval of 3.0 s, the mass of air propelled downwards by **one** propeller is 0.55 kg.

[3]

- (b) Calculate:

- (i) the increase in momentum of the mass of air in (a)

increase in momentum = N s [1]

- (ii) the downward force exerted on this mass of air by the propeller.

force = N [1]

- (c) State:

- (i) the upward force acting on **one** propeller

force = N [1]

- (ii) the name of the law that explains the relationship between the force in (b)(ii) and the force in (c)(i).

..... [1]

- (d) Determine the mass of the aircraft.

mass = kg [1]

- (e) In order for the aircraft to hover at a very high altitude (height), the propellers must propel the air downwards with a greater speed than when the aircraft hovers at a low altitude. Suggest the reason for this.

..... [1]

- (f) When the aircraft is hovering at a high altitude, an electric fault causes the propellers to stop rotating. The aircraft falls vertically downwards. When the aircraft reaches a constant speed of 22 m s^{-1} , it emits sound of frequency 3.0 kHz from an alarm. The speed of the sound in the air is 340 m s^{-1} .

Determine the frequency of the sound heard by a person standing vertically below the falling aircraft.

frequency = Hz [2]

36. M/J 17/P22/Q5

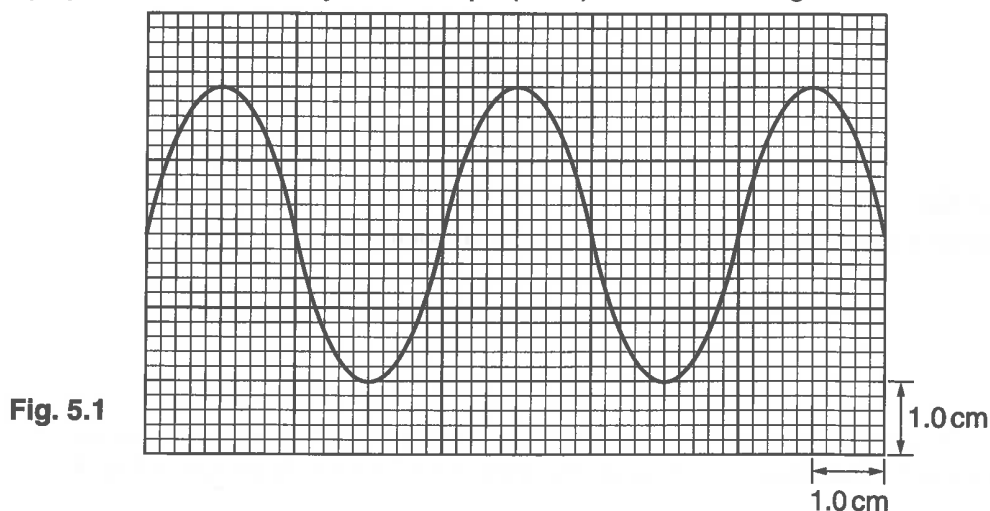
booklet (scan + OCR)

(a) Define the *frequency* of a sound wave.

..... [1]

(b) A sound wave travels through air. Describe the motion of the air particles relative to the direction of travel of the sound wave.

..... [1]

(c) The sound wave emitted from the horn of a stationary car is detected with a microphone and displayed on a cathode-ray oscilloscope (c.r.o.), as shown in Fig. 5.1.The y-axis setting is 5.0 mV cm^{-1} .The time-base setting is 0.50 ms cm^{-1} .**(i)** Use Fig. 5.1 to determine the frequency of the sound wave.

frequency = Hz [2]

(ii) The horn of the car sounds continuously. Describe the changes to the trace seen on the c.r.o. as the car travels at constant speed**1.** directly towards the stationary microphone,

.....

.....

2. directly away from the stationary microphone.

.....

..... [3]

37. M/J 17/P21/Q5

booklet (scan + OCR)

(a) Describe the Doppler effect.

[1]

(b) A car travels with a constant velocity along a straight road. The car horn with a frequency of 400 Hz is sounded continuously. A stationary observer on the roadside hears the sound from the horn at a frequency of 360 Hz.

The speed of sound is 340 m s^{-1} .

Determine the magnitude v , and the direction, of the velocity of the car relative to the observer.

$v = \dots\dots\dots \text{ms}^{-1}$

direction $\dots\dots\dots$ [3]

38. O/N 16/P21/Q4

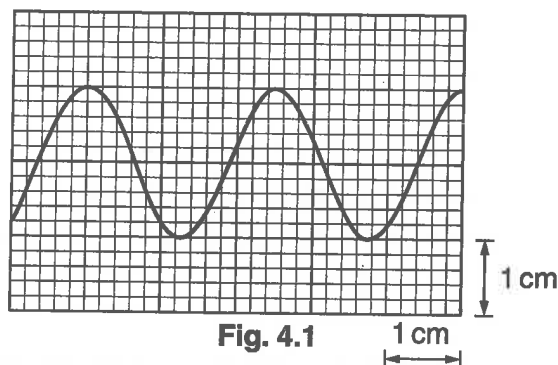
booklet (scan + OCR)

- (a) State what is meant by the *frequency* of a progressive wave.

.....

..... [2]

- (b) A cathode-ray oscilloscope (c.r.o.) is used to determine the frequency of the sound emitted by a loudspeaker. The trace produced on the screen of the c.r.o. is shown in Fig. 4.1.



The time-base setting of the c.r.o. is $250 \mu\text{s cm}^{-1}$.

Show that the frequency of the sound wave is 1600 Hz.

[2]

- (c) The loudspeaker in (b) emits the sound in all directions. A person attaches the loudspeaker to a string and then swings the loudspeaker at a constant speed in a horizontal circle above his head. An observer, standing a large distance away from the loudspeaker, hears sound of maximum frequency 1640 Hz. The speed of sound in air is 330 m s^{-1} .

- (i) Determine the speed of the loudspeaker.

speed = m s^{-1} [2]

- (ii) Describe and explain, qualitatively, the variation in the frequency of the sound heard by the observer.

.....

..... [2]

39. M/J 16/P22/Q4

booklet (scan + OCR)

- (a)** By reference to the direction of the propagation of energy, state what is meant by a *longitudinal* wave and by a *transverse* wave.

longitudinal:

transverse: [2]

- (b)** The intensity of a sound wave passing through air is given by

$$I = K v \rho f^2 A^2$$

where I is the intensity (power per unit area),

K is a constant without units,

v is the speed of sound,

ρ is the density of air,

f is the frequency of the wave

and A is the amplitude of the wave.

Show that both sides of the equation have the same SI base units.

[3]

- (c) (i)** Describe the *Doppler effect*.

..... [1]

- (ii)** A distant star is moving away from a stationary observer.

State the effect of the motion on the light observed from the star.

..... [1]

- (d)** A car travels at a constant speed towards a stationary observer. The horn of the car sounds at a frequency of 510 Hz and the observer hears a frequency of 550 Hz. The speed of sound in air is 340 m s^{-1} .

Calculate the speed of the car.

speed = m s^{-1} [3]

40. O/N 15/P21/Q5

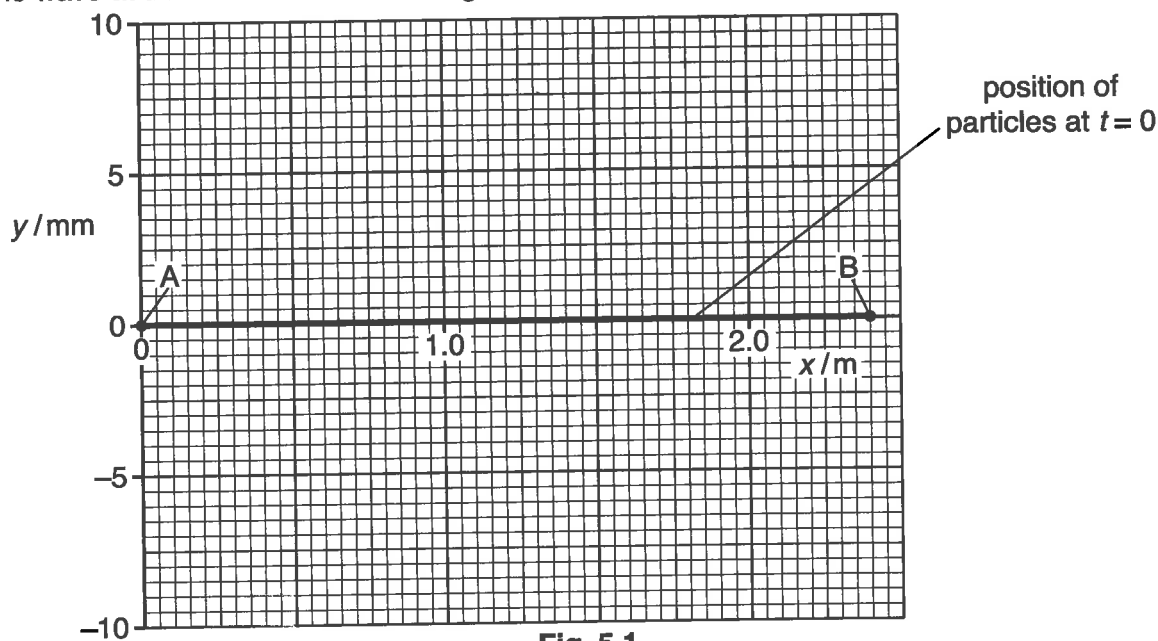
booklet (scan + OCR)

- (a) A progressive wave transfers energy. A stationary wave does not transfer energy. State two other differences between progressive waves and stationary waves.

1.

2. [2]

- (b) A stationary wave is formed on a stretched string between two fixed points A and B. The variation of the displacement y of particles of the string with distance x along the string for the wave at time $t = 0$ is shown in Fig. 5.1.

**Fig. 5.1**

The wave has a period of 20 ms and a wavelength of 1.2 m. The maximum amplitude of the particles of the string is 5.0 mm.

- (i) On Fig. 5.1, draw a line to represent the position of the string at $t = 5.0$ ms. [2]
 (ii) State the phase difference between the particles of the string at $x = 0.40$ m and at $x = 0.80$ m.

phase difference = unit [1]

- (iii) State and explain the change in the kinetic energy of a particle at an antinode between $t = 0$ and $t = 5.0$ ms. A numerical value is not required.

.....

..... [2]

41. O/N 14/P22/Q1/b

booklet (scan + OCR)

(a) An electromagnetic wave has frequency 12 THz.**(i)** Calculate the wavelength in μm .wavelength = μm [2]**(ii)** State the name of the region of the electromagnetic spectrum for this frequency.

..... [1]

42. O/N 14/P23/Q7

booklet (scan + OCR)

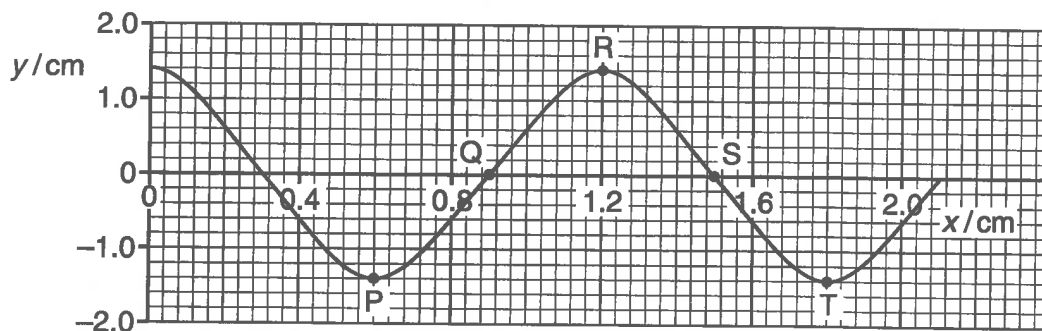
(a) (i) Explain what is meant by a *progressive transverse wave*.

progressive:

transverse: [2]

(ii) Define frequency.

..... [1]

(b) The variation with distance x of displacement y for a transverse wave is shown in Fig. 7.1.**Fig. 7.1**

On Fig. 7.1, five points are labelled.

Use Fig. 7.1 to state any two points having a phase difference of

(i) zero,

..... [1]

(ii) 270° .

..... [1]

(c) The frequency of the wave in **(b)** is 15 Hz.Calculate the speed of the wave in **(b)**.speed = ms^{-1} [3]**(d)** Two waves of the same frequency have amplitudes 1.4 cm and 2.1 cm.

Calculate the ratio

$$\frac{\text{intensity of wave of amplitude 1.4 cm}}{\text{intensity of wave of amplitude 2.1 cm}}$$

ratio = [2]

43. M/J 14/P21/Q5

booklet (scan + OCR)

(a) Explain what is meant by the following quantities for a wave on the surface of water:

(i) displacement and amplitude,

displacement

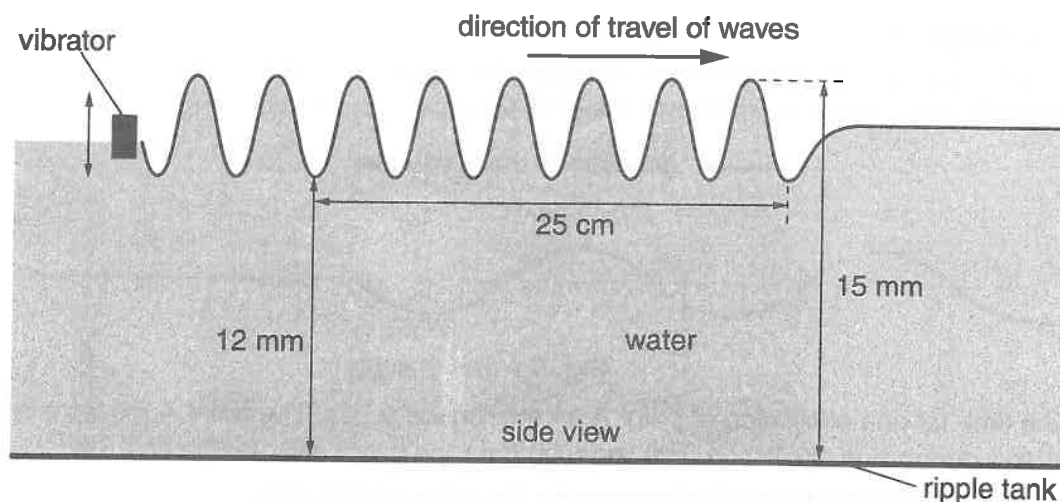
amplitude [2]

(ii) frequency and time period.

frequency

time period [2]

(b) Fig. 5.1 represents waves on the surface of water in a ripple tank at one particular instant of time.

**Fig. 5.1** (not to scale)

A vibrator moves the surface of the water to produce the waves of frequency f . The speed of the waves is 7.5 cm s^{-1} . Where the waves travel on the water surface, the maximum depth of the water is 15 mm and the minimum depth is 12 mm.

(i) Calculate, for the waves,

1. the amplitude,

amplitude = mm [1]

2. the wavelength.

wavelength = m [2]

(ii) Calculate the time period of the oscillations of the vibrator.

time period = s [2]

(c) State and explain whether the waves on the surface of the water shown in Fig. 5.1 are

(i) progressive or stationary,

..... [1]

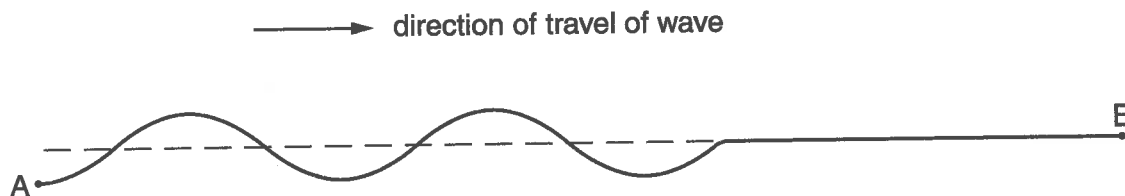
(ii) transverse or longitudinal.

..... [1]

44. O/N 13/P21/Q5

booklet (scan + OCR)

A long rope is held under tension between two points A and B. Point A is made to vibrate vertically and a wave is sent down the rope towards B as shown in Fig. 5.1.

**Fig. 5.1** (not to scale)

The time for one oscillation of point A on the rope is 0.20 s. The point A moves a distance of 80 mm during one oscillation. The wave on the rope has a wavelength of 1.5 m.

(a) (i) Explain the term *displacement* for the wave on the rope.

..... [1]

(ii) Calculate, for the wave on the rope,

1. the amplitude,

2. the speed.

amplitude = mm [1]

speed = ms^{-1} [3]

(b) On Fig. 5.1, draw the wave pattern on the rope at a time 0.050 s later than that shown. [2]

(c) State and explain whether the waves on the rope are

(i) progressive or stationary,

..... [1]

(ii) longitudinal or transverse.

..... [1]

45. O/N 13/P23/Q5

booklet (scan + OCR)

(a) (i) Define, for a wave,

1. wavelength λ ,

..... [1]

2. frequency f .

..... [1]

- (ii) Use your definitions to deduce the relationship between λ , f and the speed v of the wave.

[1]

- (b) Plane waves on the surface of water are represented by Fig. 5.1 at one particular instant of time.

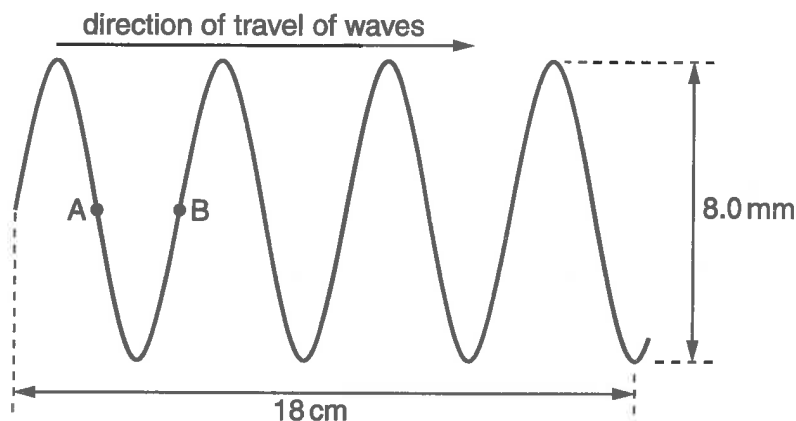


Fig. 5.1 (not to scale)

The waves have frequency 2.5 Hz.

Determine, for the waves,

- (i) the amplitude,

amplitude = mm [1]

- (ii) the speed,

speed = ms^{-1} [2]

- (iii) the phase difference between points A and B.

phase difference = unit [1]

- (c) The wave in (b) was produced in a ripple tank. Describe briefly, with the aid of a sketch diagram, how the wave may be observed.

.....

..... [2]

Answers Section

1. M/J 26/P22/Q3/a,b(i)

3(a)	(A wave which) transfers energy (in the direction of propagation)	B1	Allow 'it' to mean 'a wave' provided not clearly referring to something else Allow 'propagates / carries / moves' for transfers Ignore references to medium / matter Ignore references to transverse or longitudinal waves
3(b)(i)	$T = 0.33 \text{ s}$	A1	

2. M/J 26/P23/Q3/d

3(b)(i)	constant phase difference	B1	Ignore 'same frequency/wavelength/speed' Ignore 'have the same phase difference' Ignore 'phase relationship' for 'phase difference' Ignore 'in phase'
3(b)(ii)	(the light) diffracts	B1	Ignore 'spreads'
3(b)(iii)	waves (from each slit) meet / superpose (at the screen)	B1	Allow 'overlap / superimpose / interfere / combine' for 'superpose' Ignore 'interact / collide / coincide / intersect'
	waves have 0 path difference / 0 phase difference	B1	Allow 'in phase' Ignore 360°
	constructive interference occurs / (resultant wave has) maximum amplitude	B1	Ignore 'intensity is maximum' Ignore references to nodes and antinodes
3(b)(iv)	$\lambda = ax / D$	C1	
	$x = (560 \times 10^{-9} \times 0.43) / (1.4 \times 10^{-5})$ $= 0.017$	C1	
	distance $= 0.017 / 2$ $= 0.0086 \text{ m}$	A1	
3(d)	$c = f\lambda$	C1	Allow v for c
	frequency $= (3.00 \times 10^8) / (560 \times 10^{-9})$ $= 5.4 \times 10^{14} \text{ Hz}$	A1	ECF POT error in the wavelength from 3(b)(iv) if same POT error made

3. MAR 26/P22/Q5/a

5(a)(i)	time for one oscillation / vibration / cycle (of a particle / point on wave) or time between <u>adjacent</u> wavefronts (passing the same point) or <u>shortest</u> time between two wavefronts (passing the same point)	B1
5(a)(ii)	(The magnitude of the) maximum displacement (of a particle / point on the wave)	B1

4. O/N 25/P21/Q4/c

4(b)(iii)	peaks / maxima shown at $\theta = 0, \pm 6^\circ$ and $\pm 12^\circ$	B1
	peaks / maxima and zero intensity in between	B1
	central peak / maxima at intensity I_0 and the other peaks all equal to or less than I_0	B1
4(c)	peaks / maxima all at the same angles (as before)	B1
	intensity (of all peaks / maxima) halved	B1

5. O/N 25/P24/Q4/b(ii,iii)

4(b)(ii)	$c = f\lambda$	C1
	$f = (3.00 \times 10^8) / 0.026$	A1
	$= 1.2 \times 10^{10} \text{ Hz}$	
4(b)(iii)	microwave	B1

6. M/J 25/P21/Q5/a,b

5(a)	wavelength: - wavelength = distance between successive / adjacent in phase points / wavefronts / crests / troughs $\lambda = d / (\text{number of oscillations})$ frequency: - frequency = (number of) oscillations / cycles crests / troughs / wavefronts (passing a point) per unit time $f = (\text{number of oscillations}) / t$ One correct point from either list	B1
	One correct point from both lists and speed = distance / time and one of: - wavelength $\times$ frequency (= distance per unit time) = speed $[(\text{number of oscillations}) / t] \times [d / (\text{number of oscillations})] = f\lambda$ $v (= d / t) = \lambda / (1/f) = f\lambda$ or $v (= d / t) = \lambda / T = f\lambda$	B1
5(b)(i)	$T = 4 \times 10^{-3}$	C1
	$f = 1 / T = 1 / 0.004$	A1
5(b)(ii)	$f_o = f_s v / (v - v_s)$	C1
	$250 = 236 \times v / (v - 20)$	
	$v = (250 \times 20) / (250 - 236)$	
	$= 360 \text{ m s}^{-1}$	A1

7. M/J 25/P22/Q3/a,b(i),d

3(a)	10.0 cm	A1
3(b)(i)	$v = f\lambda$	C1
	$= 16 \times 0.40$	A1
	$= 6.4 \text{ m s}^{-1}$	
3(d)	$I \propto A^2$	C1
	$\frac{I_X}{I_Y} = 10^2 / 20^2$	
	ratio = 0.25	A1

8. M/J 25/P23/Q5/a,b

5(a)	sound waves are longitudinal / not transverse (and only transverse waves can be polarised)	B1
5(b)(i)	A periodic curve of at least one period with minimum 0 and maximum I_0	B1
	Smooth continuous $\sin^2 \alpha$ curve	B1
	Peaks at 90° and 270° and troughs at 0° , 180° and 360°	B1

5(b)(ii)	$I \propto A^2$	C1
	$I = I_0 \cos^2 \theta$	C1
	At $\alpha = 20^\circ$ the angle between the planes of polarisation of light and filter is 70° (or 110°)	C1
	$I = I_0 \cos^2 70$ or $I = I_0 \cos^2 110$	
	$(I = 0.12 I_0)$	
	amplitude = $0.34 A_0$	A1

9. M/J 25/P23/Q6/b(i)

6(b)(i)	$v = f\lambda$	C1
	$f = 3.00 \times 10^8 / 720 \times 10^{-9}$	A1
	$= 4.2 \times 10^{14} \text{ Hz}$	

10. M/J 25/P24/Q4/a,b(i)

4(a)(i)	(a wave that) transfers / propagates energy	B1
4(a)(ii)	distance = $n\lambda$	B1
	time $t = n / f$	B1
4(a)(iii)	$(v = \text{distance} / \text{time})$ $(v) = n\lambda / (n / f)$ so $(v) = f\lambda$	B1
4(b)(i)	$f = 3 \times 10^8 / 4 \times 10^{-2}$ $= 7.5 \times 10^9 \text{ (Hz)}$ $= 7.5 \times 10^9 / 10^9 \text{ (GHz)}$	C1
	$= 7.5 \text{ GHz}$	A1

11. MAR 25/P22/Q5/a,b

5(a)	$T = 5.8 \times 5.0 \times 10^{-4}$ $= 2.9 \times 10^{-3}$	C1
	$\lambda = vT$ or $v = f\lambda$ <u>and</u> $f = 1 / T$	C1
	$\lambda = 330 \times 2.9 \times 10^{-3}$ or $\lambda = 330 / 345$ $= 0.96 \text{ m}$	A1
5(b)	(loudspeaker) moves away (from the microphone)	B1
	at an increasing speed / whilst accelerating	B1

12. O/N 24/P21/Q6/b(i)

6(b)(i)	$v = f\lambda$	C1
	$\lambda = (3.00 \times 10^8) / (2.5 \times 10^{10}) = 0.012 \text{ m}$	A1

13. O/N 24/P22/Q4/a

4(a)	longitudinal waves have oscillations <u>parallel</u> to the (direction of) transfer of energy	B1
	transverse waves have oscillations <u>perpendicular</u> to the (direction of) transfer of energy	B1

14. O/N 24/P23/Q5/a(iii),b

5(a)(iii)	$v = f\lambda$	C1
	frequency = $1400 / 0.32$ = 4400 Hz	A1
5(b)(i)	$f_o = f_s v / (v - v_s)$ $f_o = (780 \times 320) / (320 - 39)$	C1
	maximum frequency = 890 Hz	A1
5(b)(ii)	line showing f varying both above and below a mean frequency and returning to the original start value of f	B1
	a single cycle of a smoothly oscillating curve of correct phase (starting at mean position, falling to a trough, then rising to a peak, ending at mean position)	B1

15. M/J 24/P21/Q5/e

5(e)	$v = f\lambda$ and $f = 1 / T$ or $v = \lambda / T$	C1
	$v = 6.25 \times 0.80$ or $0.80 / 0.16$ = 5.0 m s^{-1}	A1

16. M/J 24/P22/Q6/b(i)

6(b)(i)	(frequency of red light is) less (than frequency of original light)	B1
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17. M/J 24/P23/Q4/a,b,c

4(a)	270°	A1
4(b)	arrow pointing vertically downwards at T	A1
4(c)	$v = f\lambda$ or $v = \lambda / T$ and $f = 1 / T$	C1
	wavelength = $0.62 \times 10^{-2} \times (4 / 3)$ (= $0.83 \times 10^{-2} \text{ m}$)	C1
	$f = 0.27 / (0.83 \times 10^{-2})$ = 33 Hz	A1

18. MAR 24/P22/Q1/c

1(c)	power = intensity $\times$ area	C1
	= $950 \times 2.2 \times 10^{-4}$	C1
	= 0.21 W	A1

19. MAR 24/P22/Q5/a,b,c

5(a)	vibrations / oscillations (of the particles / wave) are perpendicular to the direction (of the propagation of energy)	B1
5(b)	infrared	B1

5(c)	$T = 6 \times 5.0 \times 10^{-15}$ $T = 3.0 \times 10^{-14}$	C1
	$\lambda = c T$ or $\lambda = c / f$ and $f = 1 / T$	C1
	$\lambda = 3.0 \times 10^8 \times 3.0 \times 10^{-14}$ or $\lambda = 3.0 \times 10^8 / 3.33 \times 10^{13}$ $= 9.0 \times 10^{-6} \text{ m}$	A1

20. O/N 23/P21/Q5/a,b

5(a)	sketch: approximately horizontal line above horizontal dashed line from $t = 0$ to $t = t_1$ and approximately horizontal line below horizontal dashed line from $t = t_1$ to $t = t_2$	A1
5(b)	$I = P / A$	C1
	$A = \pi \times 0.028^2$ or $\pi \times 2.8^2$ ($= 2.46 \times 10^{-3}$ or 24.6)	C1
	$P = 4.7 \times 10^{-3} \times 2.46 \times 10^{-3}$ $= 1.2 \times 10^{-5} \text{ W}$	A1

21. O/N 23/P22/Q5/a

5(a)(i)	light intensity has maximum value at 0° , 180° , 360° and zero intensity at 90° , 270°	M1
	'sinusoidally-shaped' curve	A1
5(a)(ii)	$4.2 = 7.6 \cos^2 \theta$	C1
	$\theta = 42^\circ$	A1

22. O/N 23/P23/Q5/b(ii)

5(b)(ii)	infrared	A1
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23. O/N 23/P23/Q6/a,b

6(a)	$291 = 251 \times 340 / (340 - v_s)$	C1
	$v_s = 47 \text{ m s}^{-1}$	A1
6(b)(i)	$I \propto A^2$	B1
6(b)(ii)	sketch: line starts at (1.0, 1.0)	B1
	approximately straight line drawn with negative gradient and line ends at (4.0, 0.25)	B1

24. M/J 23/P22/Q5

- (a) (i) (they are) perpendicular [1]
 (ii) (they are) parallel [1]
- (b) (i) $\lambda = v / f$
 $= 340 / 1700$
 $= 0.20 \text{ m}$ [2]
- (ii) $L = \frac{3}{4} \times \lambda = \frac{3}{4} \times 0.20$
 $= 0.15 \text{ m}$ [1]
- (iii) $\lambda = 4 \times 0.15$ or 0.20×3
 $= 0.60 \text{ m}$ [1]
- (c) (i) $(I =) 8.5 \times \cos^2 35^\circ = 5.7 \text{ (W m}^{-2}\text{)}$ [1]
 (ii) $5.2 = 5.7 \cos^2 \theta$
 $(\theta = 17^\circ)$
 $\alpha = 35^\circ + 17^\circ$
 $= 52^\circ$ [2]

25. M/J 23/P23/Q4

- (a) the number of wavefronts/crests/troughs passing a fixed point per unit time
 or
 the number of oscillations per unit time (of source / point on wave / particle of medium) [1]
- (b) (i) $T = 4 \times 0.50 \times 10^{-3}$
 $(= 2.0 \times 10^{-3} \text{ s})$
 $f = 1 / 2.0 \times 10^{-3}$
 $= 500 \text{ Hz}$ [2]
- (ii) amplitude $= 2.8 \times 0.20$
 $= 0.56 \text{ V}$ [1]
- (c) period same as original trace
 sinusoidal wave of constant amplitude less than 2.8 cm throughout
 amplitude 1.4 cm [3]
- (d) (i) when (two or more) waves meet (at a point)
 (resultant) displacement is the sum of the individual displacements [2]
- (ii) node-to-node separation is $\lambda / 2$
 or
 microphone moves through 3 node-to-node separations
 or
 $d = 1.5\lambda$
 $\lambda = 1.05 / 1.5$
 $= 0.70 \text{ m}$ [2]
- (iii) $v = f \times \lambda$
 $= 500 \times 0.70$
 $= 350 \text{ m s}^{-1}$ [2]

26. O/N 22/P22/Q5

- (a) (they travel in) opposite directions [1]
- (b) (i) straight line from A to B, labelled P
line that is 'mirror image' of given line, labelled Q [2]
- (ii) $\lambda = 0.80 / 2$
(= 0.40 m)
 $v = \lambda / T$
or
 $v = f\lambda$ and $f = 1 / T$
 $v = 0.40 / 0.016$ or 62.5×0.40
= 25 m s⁻¹ [3]
- (c) (i) $I_1 / I_0 = \cos^2 30^\circ$
= 0.75 [2]
- (ii) $I_2 / I_1 = \cos^2 60^\circ$
 $I_2 / I_0 = \cos^2 30^\circ \times \cos^2 60^\circ$ or $0.75 \times \cos^2 60^\circ$
= 0.19 [2]

27. O/N 22/P21/Q4

- (a) (i) oscillations are in a single direction, which is perpendicular to the direction of propagation (of the wave)
or
oscillations are in a single plane, which contains the direction of propagation (of the wave) [1]
- (ii) light waves are transverse and sound waves are longitudinal [1]
- (b) $I = I_0 \cos^2 \theta$
 $\cos^2 \theta = 1 / 4$ so $\cos \theta = 1 / 2$
 $\theta = 60^\circ$ or 120° or 240° or 300°
angle of rotation = $(120^\circ - 90^\circ)$ or $(240^\circ - 90^\circ)$ or $(300^\circ - 90^\circ)$
= 30° or 150° or 210° or 330° [3]
- (c) (i) the waves have different amplitudes
cannot have resultant displacement that is always zero
or
cannot have (complete) destructive interference (at nodes)
or
(at nodes resultant) amplitude is the difference of the amplitudes [2]
- (ii) $I \propto A^2$
 $A^2 / A_0^2 = (I_0 / 4) / I_0$
 $A = 0.5 A_0$
maximum amplitude = $A_0 + 0.5 A_0$
= $1.5 A_0$ [3]

28. O/N 22/P23/Q4

- (a) (i) circles drawn around any two particles with seven (uncircled) particles in between [1]
- (ii) curve has an initial negative displacement and initial amplitude same as original curve
curve has same amplitude as original curve throughout
curve has same wavelength as original curve throughout, with constant (non-zero) phase difference [3]

- (iii) (to the) right / rightwards [1]
- (b) $\lambda = 2 \times 0.19$
 $= 0.38 \text{ m}$
 $v = f\lambda$
 $v = (16 \times 10^3) \times 0.38$
 $v = 6100 \text{ m s}^{-1}$ [3]
- (c) $I \propto A^2$
 $I = (1/2)^2 I_0$
 intensity = $0.25 I_0$ [2]
- (d) (i) same frequency / wavelength / period [1]
- (ii) • a stationary wave has nodes/antinodes (and a progressive wave does not)
 • a stationary wave does not transfer/propagate energy (and a progressive wave does transfer/propagate energy)
 • different points on a stationary wave have different amplitudes (and all points on a progressive wave have the same/constant amplitude)
 • stationary wave has adjacent particles that are in phase (and adjacent particles on progressive wave are out of phase)
- Any two points, 1 mark each. Allow the reverse statement for each marking point.* [2]

29. M/J 22/P21/Q5

- (a) wave(s) (travel along string and) reflect at fixed point / A / B / end
 incident and reflected waves superpose / interfere
 or
 two waves travelling / with speed in opposite directions superpose / interfere [2]
- (b) line has an approximate sinusoidal shape with maximum downward displacement at P
 and zero displacement at each node [1]
- (c) $v = \lambda / T$
 or
 $v = f\lambda$ and $f = 1 / T$
 $\lambda = 35 \times 0.040$ or $35 / 25$
 (= 1.4 m)
 distance = 1.4×2.5
 = 3.5 m [3]
- (d) (number of periods / cycles) ($= t / T$) = $0.060 / 0.040$
 (= 1.5)
 amplitude = $72 / 6$
 = 12 mm [2]

30. O/N 21/P22/Q4

- (a) (i) decrease(s) [1]
 (ii) increase(s) [1]
- (b) $f_o = f_s v / (v + v_s)$
 $= 925 \times 338 / (338 + 12) = 893 \text{ Hz}$ [2]
- (c) distance = $(\frac{1}{2} \times 2 \times 12) + (2 \times 12)$ (= 36 m)
 time taken = $36 / 338 = 0.11 \text{ s}$ [2]

31. O/N 21/P21/Q4

- (a) oscillations (of particles) are parallel to (the direction of) energy transfer [1]

- (b) (i) (frequency varies as) vehicle moves relative to (stationary) observer (vehicle) moving towards (observer) gives higher (observed) frequency (than 1.2 kHz) and (vehicle) moving away (from observer) gives lower (observed) frequency (than 1.2 kHz) [2]
 (ii) Doppler effect [1]
 (iii) position of vehicle labelled 'X' at top (12 o'clock) position on track [1]
 (iv) position of vehicle labelled 'Y' at right-hand edge (3 o'clock) position on track [1]
 (c) maximum frequency = 1.40 (kHz) **or** 1.40×10^3 (Hz)
 $1.40 = (1.2 \times 320) / (320 - v)$
 $v = 46 \text{ m s}^{-1}$ **or** minimum frequency = 1.05 (kHz) **or** 1.05×10^3 (Hz)
 $1.05 = (1.2 \times 320) / (320 + v)$
 $v = 46 \text{ m s}^{-1}$ [3]

32. MAR 21/P22/Q5

- (a) $f_o = f_s v / (v + v_s)$
 $f_o = 951 \times 330 / (330 + 12) = 918 \text{ Hz}$ [2]
 (b) $t = d / 12$
 $= (\pi \times 2.4) / 12 = 0.63 \text{ s}$ [2]

33. O/N 20/P21/Q5

- (a) $v = \lambda / T$ **or**
 $v = f\lambda$ **and** $f = 1 / T$
 $v = 8.0 \times 10^{-2} / 0.40 = 0.20 \text{ m s}^{-1}$ [2]
 (b) $I \propto A^2$
 ratio = $2^2 / 4^2 = 0.25$ [3]

34. M/J 20/P23/Q4

- (a) (i) frequency **or** period [1]
 (ii) amplitude [1]
 (b) constant phase difference so coherent [1]
 (c) 120° [1]
 (d) resultant displacement = $4.0 \mu\text{m} - 1.0 \mu\text{m} = 3.0 \mu\text{m}$ [1]
 (e) $I \propto A^2$
 intensity of Z = $(2^2 / 4^2) I = 0.25 I$ [2]
 (f) $v = \lambda / T$ **or**
 $v = f\lambda$ **and** $f = 1 / T$
 $330 = \lambda / 3.0 \times 10^{-3}$
 $\lambda = 0.99 \text{ m}$ [3]

35. O/N 19/P21/Q3

- (a) $\rho = m / V$
 $V = \pi \times (0.16 / 2)^2 \times 7.6 \times 3.0 (= 0.458 \text{ m}^3)$
 $m = \pi \times (0.16 / 2)^2 \times 7.6 \times 3.0 \times 1.2 = 0.55 \text{ kg}$ [3]
 (b) (i) $\Delta p = 0.55 \times 7.6 = 4.2 \text{ N s}$ [1]
 (ii) $F = 4.2 / 3.0$ **or** $0.55 \times 7.6 / 3.0 = 1.4 \text{ N}$ [1]
 (c) (i) $F = 1.4 \text{ N}$ [1]
 (ii) Newton's third law (of motion) [1]
 (d) $2 \times 1.4 = m \times 9.81$
 $m = 0.29 \text{ kg}$ [1]
 (e) the density of air is less at high altitude [1]
 (f) $f_o = f_s v / (v - v_s)$
 $= 3000 \times 340 / (340 - 22) = 3200 \text{ Hz}$ [2]

36. M/J 17/P22/Q5

- (a) frequency is the number of vibrations/oscillations per unit time or the number of wavefronts passing a point per unit time [1]
- (b) vibrations/oscillation of the air particles are parallel to the direction of it (the direction of travel of the sound wave) [1]
- (c) (i) $T = 2(.0)$ (ms)
 $f = 500$ Hz [2]
- (ii) 1. amplitude increases
 (time) period decreases
 2. amplitude decreases
 (time) period increases [3]

37. M/J 17/P21/Q5

- (a) observed frequency is different to source frequency when source moves relative to observer
- (b) $360 = (400 \times 340) / (340 \pm v)$ [1]
 $v = 38$ (37.8) m s^{-1}
 away (from the observer) [3]

38. O/N 16/P21/Q4

- (a) the number of oscillations per unit time of the source/of a point on the wave/of a particle (in the medium) or the number of wavelengths/wavefronts per unit time passing a (fixed) point [2]
- (b) T or period = 2.5×250 (μs) (= 625 μs)
 frequency = $1 / (6.25 \times 10^{-4})$ or $1 / (2.5 \times 250 \times 10^{-6}) = 1600$ Hz [2]
- (c) (i) for maximum frequency: $f_o = f_s v / (v - v_s)$
 $1640 = (1600 \times 330) / (330 - v_s)$
 $v_s = 8(.0) \text{ m s}^{-1}$ (8.049 m s^{-1}) [2]
- (ii) loudspeaker moving towards observer causes rise in/higher frequency
 loudspeaker moving away from observer causes fall in/lower frequency or repeated rise and fall/higher and then lower frequency caused by loudspeaker moving towards and away from observer [2]

39. M/J 16/P22/Q4

- (a) longitudinal: vibrations/oscillations (of the particles/wave) are parallel to the direction or in the same direction (of the propagation of energy)
 transverse: vibrations/oscillations (of the particles/wave) are perpendicular to the direction (of the propagation of energy) [2]
- (b) LHS: intensity = power / area units: $\text{kg m s}^{-2} \times \text{m} \times \text{s}^{-1} \times \text{m}^{-2}$ or $\text{kg m}^2 \text{s}^{-3} \times \text{m}^{-2}$
 RHS: units: $\text{m s}^{-1} \times \text{kg m}^{-3} \times \text{s}^{-2} \times \text{m}^2$
 LHS and RHS both kg s^{-3} [3]
- (c) (i) change/difference in the observed/apparent frequency when the source is moving (relative to the observer) [1]
 (ii) wavelength increases/frequency decreases/red shift [1]
- (d) observed frequency = $v f_s / (v - v_s)$
 $550 = (340 \times 510) / (340 - v_s)$
 $v_s = 25$ (24.7) m s^{-1} [3]

40. O/N 15/P21/Q5

- (a) progressive: all particles have same amplitude
 stationary: no nodes or antinodes or maximum to minimum/zero amplitude

progressive: adjacent particles are not in phase

stationary: waves particles are in phase (between adjacent nodes)

[2]

(b) (i) wavelength 1.2 m (zero displacement at 0.0, 0.60 m, 1.2 m, 1.8 m, 2.4 m)

either peaks at 0.30 m and 1.5 m and troughs at 0.90 m and 2.1 m
or vice versa (but not both)

maximum amplitude 5.0 mm

[2]

(ii) 180° or π rad

[1]

(iii) at $t = 0$ particle has kinetic energy as particle is moving

at $t = 5.0$ ms no kinetic energy as particle is stationary

so decrease in kinetic energy (between $t = 0$ and $t = 5.0$ ms)

[2]

41. O/N 14/P22/Q1/b

(a) (i) wavelength = $3 \times 10^8 / 12 \times 10^{12}$
= $25 \mu\text{m}$

[2]

(ii) infra-red / IR

[1]

42. O/N 14/P23/Q7

(a) (i) progressive: energy is moved / transferred / propagated from one place to another (without the bulk movement of the medium)

transverse: (particles) oscillate / vibrate at right angles to the direction of travel of the energy / wavefront

[2]

(ii) number of oscillations per unit time / number of wavefronts passing a point per unit time

[1]

(b) (i) P and T

[1]

(ii) P and S or Q and T

[1]

(c) $\lambda = 1.2 \times 10^{-2}$ (m)

$$v = f\lambda$$

$$= 15 \times 1.2 \times 10^{-2}$$

$$= 0.18 \text{ m s}^{-1}$$

[3]

(d) ratio = $(1.4)^2 / (2.1)^2$
= 0.44

[2]

43. M/J 14/P21/Q5

(a) (i) displacement is the distance from the equilibrium position / undisturbed position / midpoint / rest position
amplitude is the maximum displacement

[2]

(ii) frequency is the number of wavefronts / crests passing a point per unit time / number of oscillations per unit time
time period is the time between adjacent wavefronts
or time for one oscillation

[2]

(b) (i) 1. amplitude = 1.5 mm

[1]

2. wavelength = $25/6$

$$= 4.2 \text{ cm or } 4.2 \times 10^{-2} \text{ m}$$

[2]

(ii) $v = \lambda / T$ or $v = f\lambda$ and $T = 1/f$

$$T = 4.2 / 7.5 = 0.56 \text{ s}$$

[2]

- (c) (i) progressive
wavefront/crests moving/energy is transferred by the waves [1]
- (ii) transverse
the vibration is perpendicular to the direction of energy transfer/wave velocity
or travel of the wave/wavefronts [1]

44. O/N 13/P21/Q5

- (a) (i) displacement is the distance the rope / particles are (above or below) from the equilibrium / mean / rest / undisturbed position (not 'distance moved') [1]
- (ii) 1. amplitude $(= 80 / 4) = 20 \text{ mm}$ [1]
2. $v = f\lambda$ or $v = \lambda / T$
 $f = 1 / T = 1 / 0.2 \text{ (5 Hz)}$
 $v = 5 \times 1.5 = 7.5 \text{ m s}^{-1}$ [3]
- (b) point A of rope shown at equilibrium position
same wavelength, shape, peaks / wave moved $\frac{1}{4}\lambda$ to right [2]
- (c) (i) progressive as energy OR peaks OR troughs is/are transferred/moved /propagated (by the waves) [1]
- (ii) transverse as particles/rope movement is perpendicular to direction of travel /propagation of the energy/wave velocity [1]

45. O/N 13/P23/Q5

- (a) (i) 1. wavelength: minimum distance between two points moving in phase
OR distance between neighbouring or consecutive peaks or troughs
OR wavelength is the distance moved by a wavefront in time T or one oscillation/cycle or period (of source) [1]
2. frequency: number of wavefronts / (unit) time
OR number of oscillations per unit time or oscillations/time [1]
- (ii) speed = distance / time = wavelength / time period
 $= \lambda / T = \lambda f$ [1]
- (b) (i) amplitude = 4.0 mm (allow 1 s.f.) [1]
- (ii) wavelength = $18 / 3.75 (= 4.8)$
speed = $2.5 \times 4.8 \times 10^{-2} = 12 \times 10^{-2} \text{ m s}^{-1}$ unit consistent with numerical answer, e.g. in cm s^{-1} if cm used for λ and unit changed on answer line [2]
[if $18 \text{ cm} = 3.5\lambda$ used giving speed 13 (12.9) cm s^{-1} allow max. 1].
- (iii) 180° or $\pi \text{ rad}$ [1]
- (c) light and screen and correct positions above and below ripple tank
strobe or video camera [2]